

Course in

Seismic processing for interpreters

Tutorial, practice and exercises
Statoil Seismic Processing Network

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Introduction to exercises:

Introduction to ProMAX

Course in
Seismic processing for interpreters

Intro to ProMAX and its User Interface

What is ProMAX?

Landmark tool to create, modify, and execute seismic processing flows.

Interactive tool with User Interface primarily mouse-driven.

Runs on Sun Solaris, Silicon Graphics and Linux workstations

ProMAX Overview

ProMAX breaks groups of datasets and processing parameters into

Areas

Lines

Flows

Areas contain one or more Line(s)

Lines contain one or several Flow(s)

A Flow is a sequence of processes to manipulate seismic data

A Typical ProMAX Flow...

...contains a data input process, one or several data manipulation processes, and a display/output process.

Intro to ProMAX and its User Interface

What can ProMAX do?

- Process entire seismic surveys:
 - Marine and Land
 - 2D, 3D, VSP
- New functionality includes tools for
 - 4D-processing
 - 3D Visualization (GeoProbe link)
- Limited functionality for processing multi-component seismic data (OBC)

Who can use ProMAX?

Anyone with a valid unix user-id can apply for access via "[Access IT](#)" on IntraNet

ProMAX is installed in Stavanger, Trondheim, Stjørdal, Harstad and Bergen

Intro to ProMAX and its User Interface

Flows are built by bringing in processes from the Processes List Window (MB1)

Processes in a flow are executed in their sequential order.

Area Line Flow

The screenshot displays the ProMAX 2D software interface. The top status bar indicates 'ProMAX 2D Version 2003.3.3 © Landmark Graphics Corporation 1989-2002. All rights Reserved. MID : 000bcd42cc6f'. The main window is titled 'AREA: Training for interpreters' and shows a flow editor. The flow is named 'LINE: Team A' and contains a single process '01) Load SEG-Y Seismic Data'. The 'Flow Editing Window' is highlighted with a red box, showing the process list and the 'Add' button. The 'Processes List Window' is also highlighted with a red box, showing a list of available processes. The 'Area Line Flow' text is written in a stylized font on the left side of the image.

Flow Editing Window

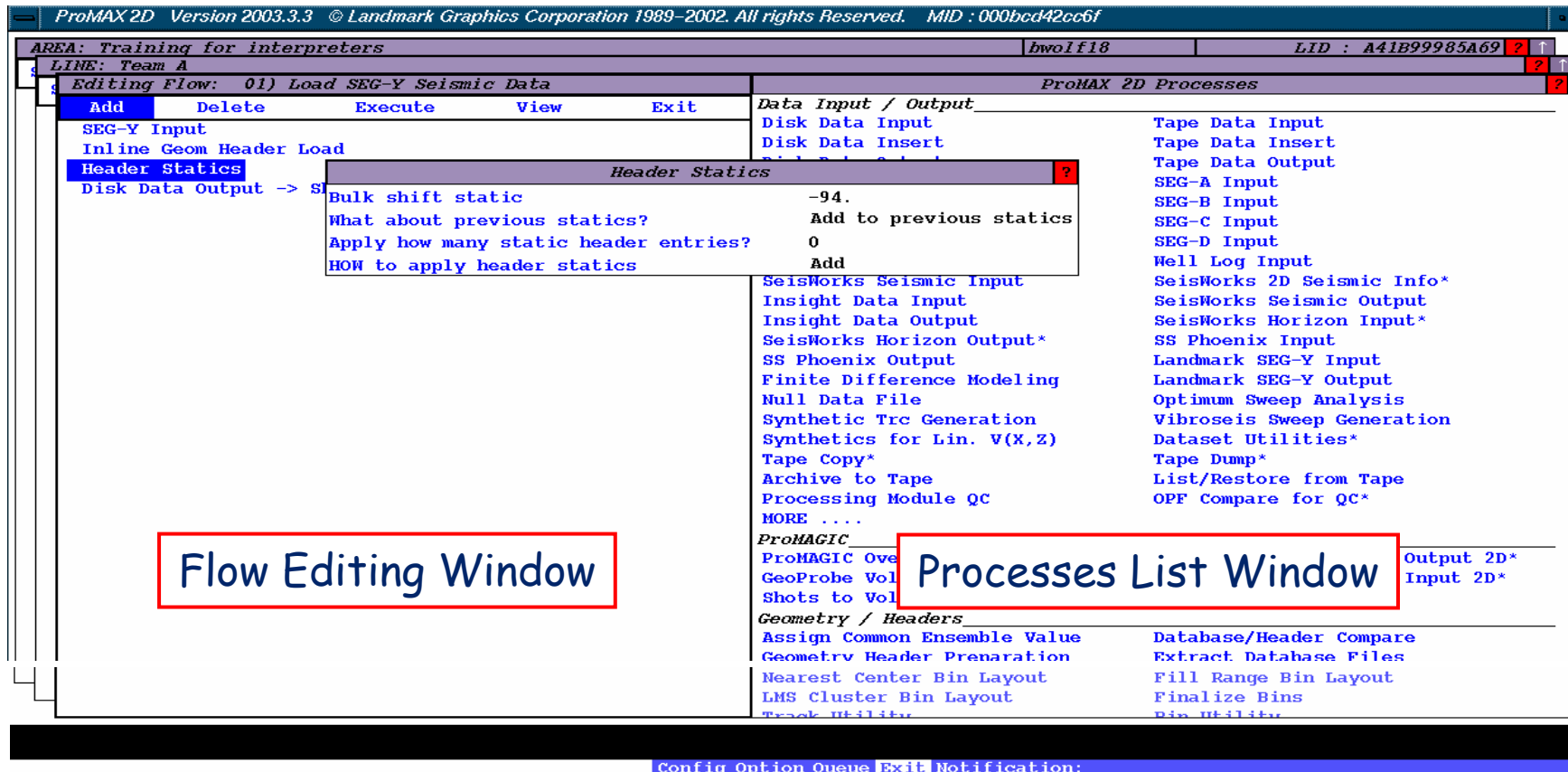
Processes List Window

Config Option Queue Exit Notification:

Intro to ProMAX and its User Interface

Most ProMAX processes have several associated internal parameters

View/Edit these process parameters by clicking MB2 on process



Intro to ProMAX and its User Interface

Online Help is available as Mouse-button help messages on lower left...

...and as PDF help files for the individual Processes

Details in Chapter 10
in the Course manual

The screenshot displays the ProMAX 2D software interface. At the top, a status bar shows "ProMAX 2D Version 2003.3.3 © Landmark Graphics Corporation 1989-2002. All rights Reserved. MID : 000bcd42cc6f". Below this, a menu bar includes "AREA: Training for interpreters", "LINE: Team A", "bwo1f18", and "LID : A41B99985A69". The main window is divided into several panes. On the left, the "Flow Editing Window" shows a list of processes under the heading "Editing Flow: 01) Load SEG-Y Seismic Data". The central pane displays the "Header Statics" dialog box, which includes fields for "Bulk shift static" (set to -94), "What about previous statics?" (set to "Add to previous statics"), "Apply how many static header entries?" (set to 0), and "HOW to apply header statics" (set to "Add"). On the right, the "Processes List Window" shows a list of processes, including "Tape Data Input", "Tape Data Insert", "Tape Data Output", "SEG-A Input", "SEG-B Input", "SEG-C Input", "SEG-D Input", "Landmark SEG-Y Output", "Optimum Sweep Analysis", "Vibroseis Sweep Generation", "Dataset Utilities*", "Tape Dump*", "List/Restore from Tape", "OPF Compare for QC*", "GeoProbe Surface File Output 2D*", "GeoProbe Surface File Input 2D*", "Shots to Vol", "Geometry / Header", "Assign Common", "Geometry Header Preparation", "Nearest Center Bin Layout", "LMS Cluster Bin Layout", "Track Utility", "Extract Database Files", "Fill Range Bin Layout", "Finalize Bins", and "Bin Utility". At the bottom left, a "Mouse-button help messages" window is visible. At the bottom right, a "View Online PDF Help Files" button is shown. The bottom status bar displays "Config Option Queue Exit Notification:".

ProMAX 2D Version 2003.3.3 © Landmark Graphics Corporation 1989-2002. All rights Reserved. MID : 000bcd42cc6f

AREA: Training for interpreters | bwo1f18 | LID : A41B99985A69

LINE: Team A

Editing Flow: 01) Load SEG-Y Seismic Data

ProMAX 2D Processes

Flow Editing Window

Processes List Window

Mouse-button help messages

View Online PDF Help Files

Config Option Queue Exit Notification:

Starting up the ProMAX User Interface

Traditionally ProMAX has been (and still is) run on Solaris and IRIX.

We will in this course run **ProMAX on Linux**

The cluster contains 36 dual-CPU processing nodes,
each with a unique name: **bwolf1, bwolf2.....bwolf36**
Nodes are chosen based on current cluster load.



Step 1

Log on to your assigned UNIX workstation using your own user-id.

Step 2

Open up a window from your workstation to your assigned Linux node.

For nodes 1-20: "rlogin bwolf16" (set DISPLAY manually)
For nodes 21-36: "xlogin bwolf21" (DISPLAY automatically set)

Step 3

Start the ProMAX User Interface from the unix prompt by typing in:

% promax (Full path in Stavanger is '/prog/promax/script/promax')

Introduction to exercises:

Loading and Viewing Data

Course in
Seismic processing for interpreters

Exercise: Loading and Viewing Data

In this exercise:

- Build two flows that will
 - Read in seismic shots records
 - Load geometry to trace headers
 - Correct for timing delay in recording
 - Write data to ProMAX internal data set
 - View CDP-gathers for quick data quality control

General data loading in ProMAX:

- From seismic tapes or disk files (SEG-Y or SEG-D)
- From OpenWorks database (usually post-stack data)

In this particular processing course:

- From seismic disk image SEG-Y file

Exercise: Loading and Viewing Data

AREA: Training for interpreters		bwolf28		LID : A41B99985A	
LINE: Team A					
Editing Flow: 01) Load SEG-Y Seismic Data		ProMAX 2D Processes			
Add	Delete	Execute	View	Exit	
SEG-Y Input		SEG-Y Input			
Inline Geom Header Load		Type of SEG-Y			
Header Statics		Type of storage to use			
Disk Data Output -> SH		Enter DISK file path name			
		Update LIN database at end of input?			
		Override input data's sample interval?			
		Samples per data trace (override binary header)			
		Store reel header in processing history?			
		Input AUXILIARY traces?			
		Get CHANNEL NUMBER from trace headers?			
		Input trace FORMAT			
		Apply trace weighting factors (2**-N)?			
		Display ensemble information?			
		Maximum TIME to input			
		Is this STACKED data?			
		MAX traces per ensemble			
		Primary SORT header word (domain of data)			
		Input PRIMARY selection choice?			
		Input SECONDARY selection choice?			
		Input Global XY reference coordinates?			
		Use the coordinate scalar?			
		Remap SEG-Y header values?			
		Standard fixed trace length			
		Disk Image			
		/private/eobre/data/snohvit_shot			
		Yes No			
		Yes No			
		0			
		Yes No			
		Yes No			
		Yes No			
		Get from header			
		Yes No			
		Yes No			
		0.			
		Yes No			
		120			
		SHOT			
		Input ALL			
		None			
		Yes No			
		Yes No			
		Yes No			
		Geometry / Headers			
		Assign Common Ensemble Value			
		Database			

Reading seismic shot records from disk:

- Specify disk image storage
- Disk file path name:

/private/eobre/data/snohvit_shots.segy

Exercise: Loading and Viewing Data

M/V ZEPHYR 1		COVER PAGE	
<u>MARINE SEISMIC OBSERVERS LOG</u>			
Client:	TGS NOPEC	LINE NUMBER:	BSS01- 106
Contractor:	DMNG	SEQUENCE:	12

RECORDING DETAILS		SINGLE STREAMER DETAILS	
Hi cut	218Hz/484db	Streamer type	SYNTRAK RDA
Lo cut	3Hz/6db	Number of groups	480 traces
Sample System type	Syntrak 480	Group length	12.5 meters
Filter rate	2milliseconds	Number of phones per group	16
Record length	8 seconds	Target streamer depth	8 meters+/-1m
Tape format	SEG D 8015	Near trace	480 traces
Tape cartridge	3590/10GB	Shotpoint interval	25 meters
A/D & Filter delay	34 milliseconds	Near trace offset	144m
Extended header	YES	SOURCE DETAILS	
Record delay	NONE	Source	Tuned Bolt array
Total recording static shift	94 milliseconds	Source size	2940 cubic inch
Gain mode:	Automatic gain control	Source depth	7 meters
		Air pressure	2000 psi
		Gun delay	60 milliseconds

Correcting for recording system timing delay:

AREA: Training for interpreters		bwolf28	
LINE: Team A			
Editing Flow: 01) Load SEG-Y Seismic Data		ProMAX 2D Proc	
Add	Delete	Execute	View
SEG-Y Input		Data Input / Output	
Header Statics		Disk Data Input	
Inline Geom Header Load		Bulk shift static	
Disk Data Output -> II		-94.	
		What about previous statics?	
		Add to previous statics	
		Apply how many static header entries?	
		0	
		HOW to apply header statics	
		Add	
		Floppy Input	
		Well	

Exercise: Loading and Viewing Data

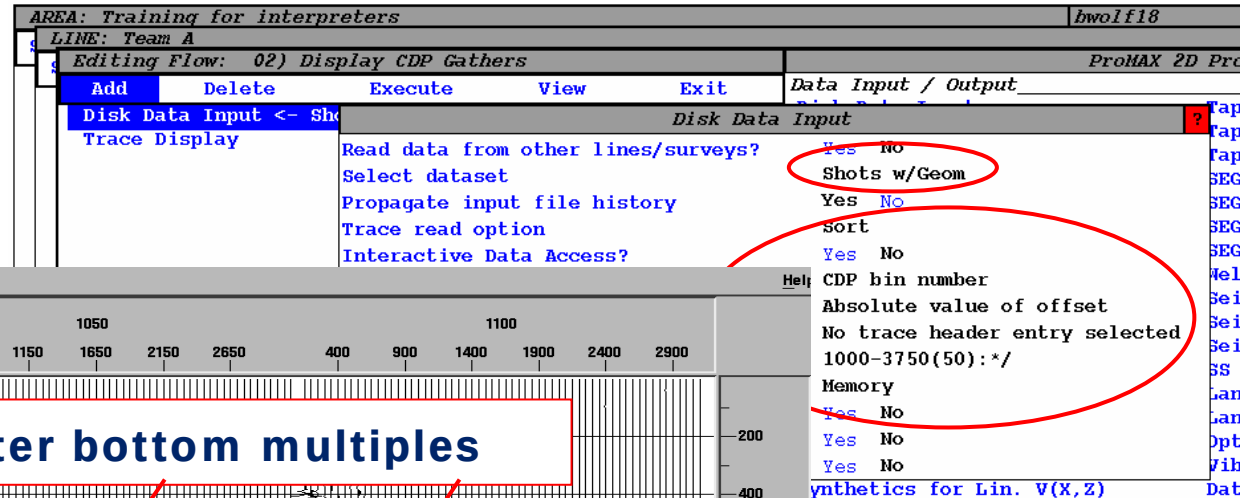
Loading CDP-gathers

AREA: Training for interpreters		hwolf18
LINE: Team A		
Editing Flow: 02) Display CDP Gathers		ProMAX 2D Pro
Add	Delete	Execute View Exit
Disk Data Input <- Sh		Disk Data Input
Trace Display		
Read data from other lines/surveys?		Yes No
Select dataset		Shots w/Geom
Propagate input file history		Yes No
Trace read option		Sort
Interactive Data Access?		Yes No
Select primary trace header entry		CDP bin number
Select secondary trace header entry		Absolute value of offset
Select tertiary trace header entry		No trace header entry selected
Sort order list for dataset		1000-3750(50):*/
Presort in memory or on disk?		Memory
Read the data multiple times?		Yes No
Process trace headers only?		Yes No
Override input data's sample interval?		Yes No
		Synthetics for Lin. V(X,Z)

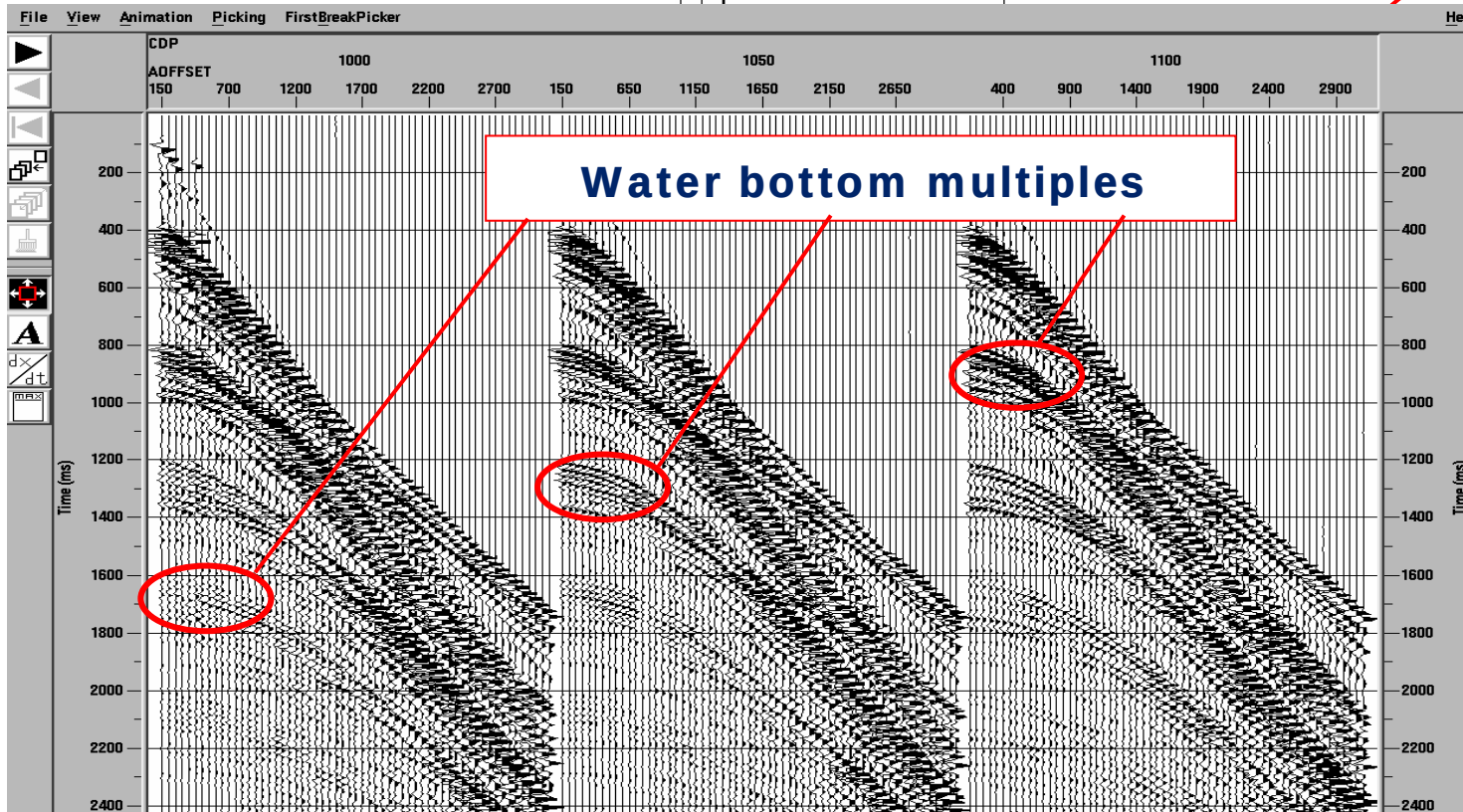
- Load *Shots w/Geom* data set
- Sort input data based into CDP-domain

Exercise: Loading and Viewing Data

Loading CDP-gathers



Water bottom multiples



Viewing CDP-gathers

Introduction to exercises:

Brute Stack

Course in

Seismic processing for interpreters

Exercise: Brute Stack

In this exercise:

Build a ProMAX flow that will

- Read in our seismic CDP gathers
- Correct for Normal Moveout (NMO)
- Apply a top mute
- Stack and display the section

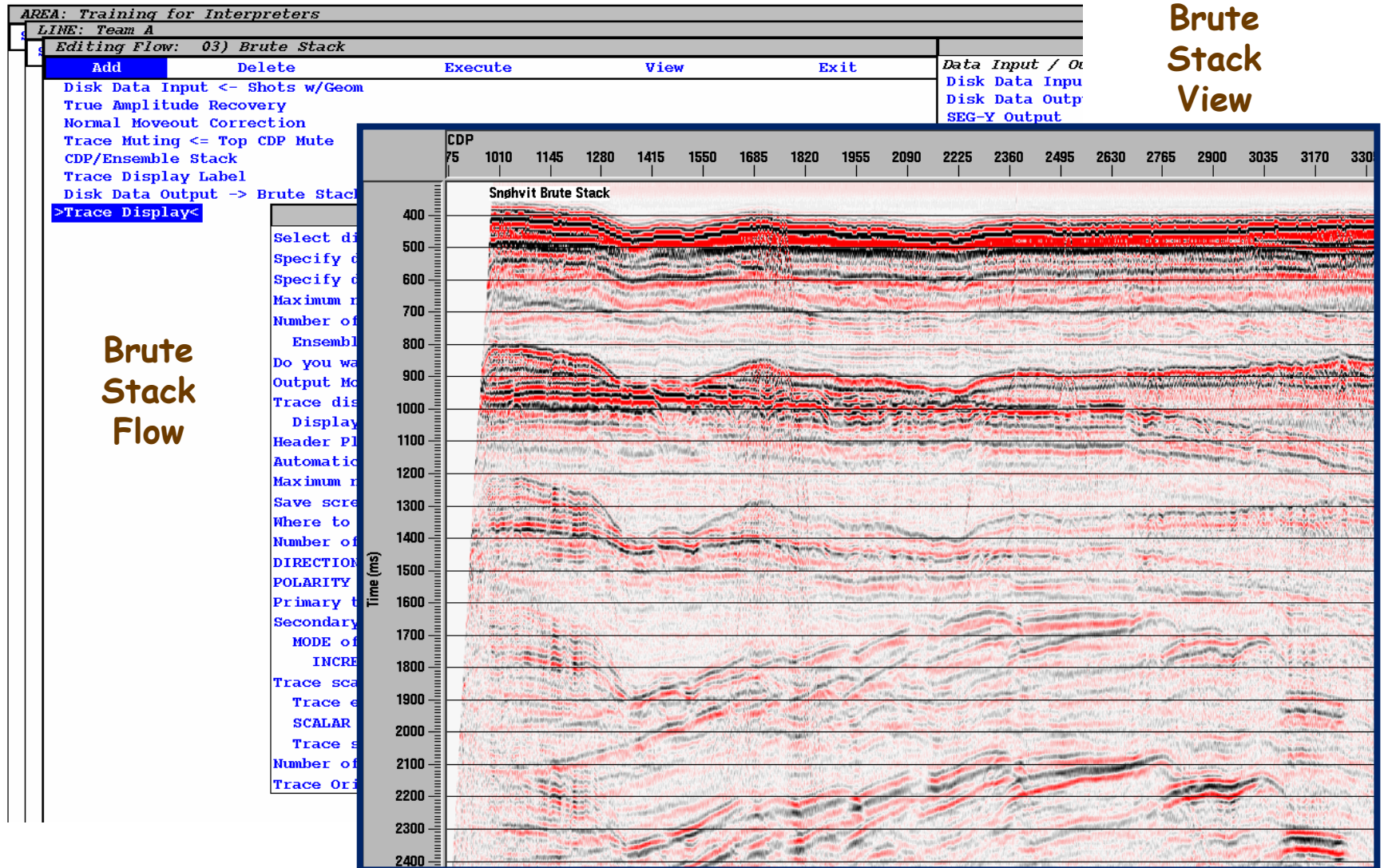
Exercise: Brute Stack

AREA: Training for Interpreters	
LINE: Team A	
Editing Flow: 03) Brute Stack	
Add	Delete
Disk Data Input <- Shots w/Geom	
True Amplitude Recovery	
Normal Moveout Correction	
Trace Muting <= Top CDP Mute	
CDP/Ensemble Stack	
Trace Display Label	
Disk Data Output -> Brute Stack	
>Trace Display<	

Trace Display	
Select display DEVICE	This Screen
Specify display START time	0.
Specify display END time	2750.
Maximum number of TRACES/screen	0
Number of ENSEMBLES(line segments)/screen	1
Ensemble increment	0
Do you want to use variable trace spacing?	Yes No
Output Mode	When Done
Trace display MODE	Color
Display color bar?	Yes No
Header Plot Parameter	No trace header selected
Automatically SAVE screens?	Yes No
Maximum number of screen images to save.	10
Save screens in Color?	Yes No
Where to save screen images	Xserver
Number of screens to collect	1
DIRECTION of trace plotting	Left to right
POLARITY of trace display	Normal
Primary trace LABELING header entry	No trace header entry selected
Secondary trace LABELING header entry	CDP bin number
MODE of Secondary trace annotation	Incremental
INCREMENT for Secondary trace annotation	5
Trace scaling mode	Conventional
Trace excursion at which to CLIP	2.
SCALAR for sample value multiplication	0.2
Trace scaling option	No scaling
Number of display panels	1
Trace Orientation	Vertical

Brute
Stack
Flow

Exercise: Brute Stack



Introduction to exercises:

BandPass Filters & Frequency

Course in
Seismic processing for interpreters

Exercise: BandPass Filters & Frequency

In this exercise:

Build a ProMAX flow that will

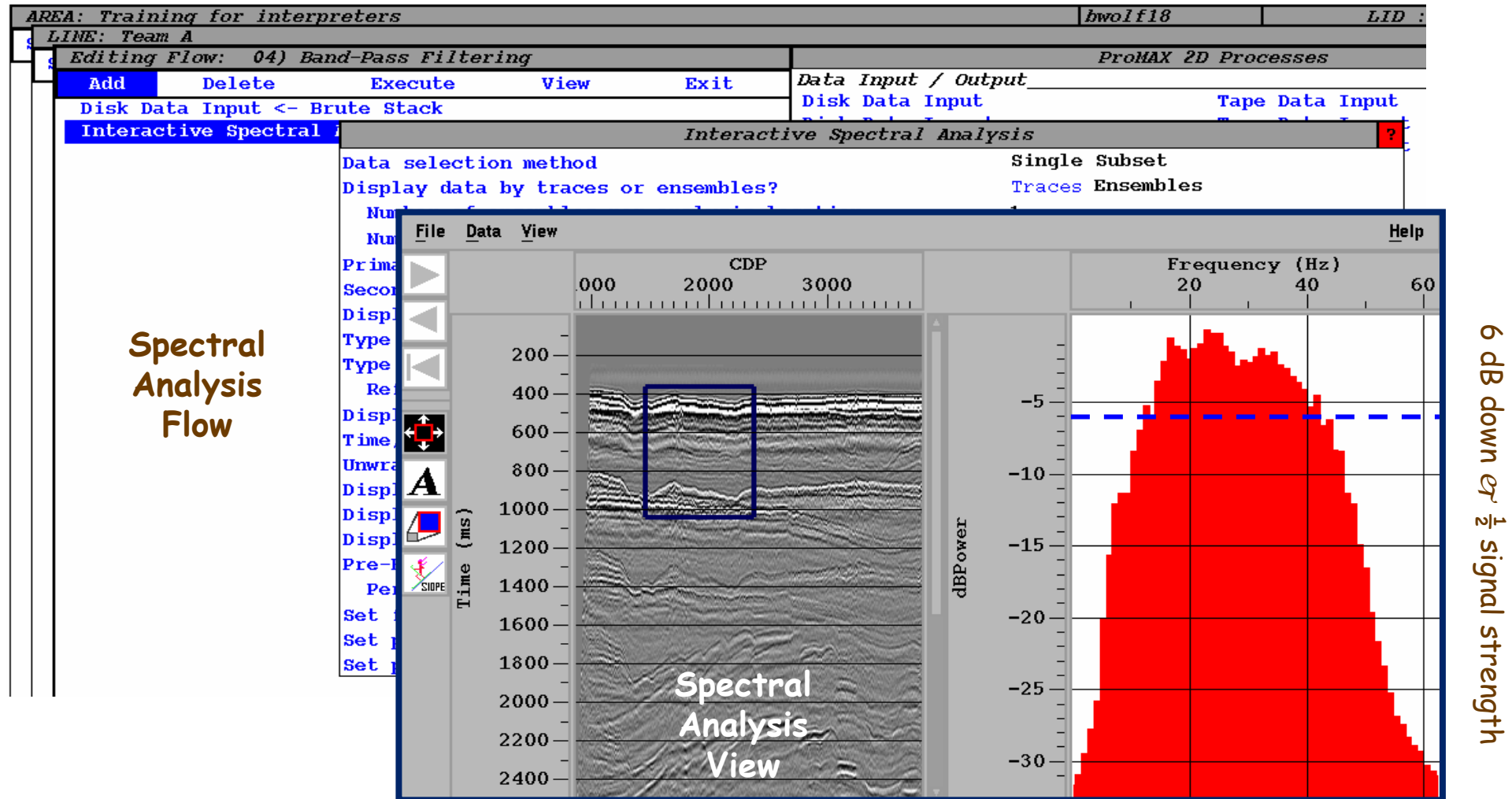
- Interactively analyze brute stack frequency content
- Apply and see effect of band-pass filters on the data

Exercise: BandPass Filters & Frequency

AREA: Training for interpreters		bwo1f18		LID :	
LINE: Team A					
Editing Flow: 04) Band-Pass Filtering		ProMAX 2D Processes			
Add	Delete	Execute	View	Exit	
Disk Data Input <- Brute Stack		Disk Data Input		Tape Data Input	
Interactive Spectral		Interactive Spectral Analysis			
		Data selection method			
		Single Subset			
		Display data by traces or ensembles?			
		Traces Ensembles			
		Number of ensembles per analysis location			
		1			
		Number of ensembles between analysis locations			
		1			
		Primary header for sorting and trace label			
		CDP bin number			
		Secondary header for sorting and trace label			
		No trace header entry selected			
		Display the average power spectrum?			
		Yes No			
		Type of scaling for power spectrum			
		dB Power			
		Type of mapping for power spectrum			
		Linear			
		Reference power			
		-1.			
		Display the average phase spectrum?			
		Yes No			
		Time, sample, linear or no phase shift?			
		None			
		Unwrap the phase spectrum?			
		Yes No			
		Display the selected trace data?			
		Yes No			
		Display the FX power spectrum?			
		Yes No			
		Display the pre-FFT time window?			
		Yes No			
		Pre-FFT time window taper type			
		Hanning			
		Percent flat for the pre-FFT time window			
		80.			
		Set frequency display ranges automatically?			
		Yes No			
		Set power display ranges automatically?			
		Yes No			
		Set phase display ranges automatically?			
		Yes No			
		Assign Common Ensemble Value		Database/Header Co	

Spectral
Analysis
Flow

Exercise: BandPass Filters & Frequency



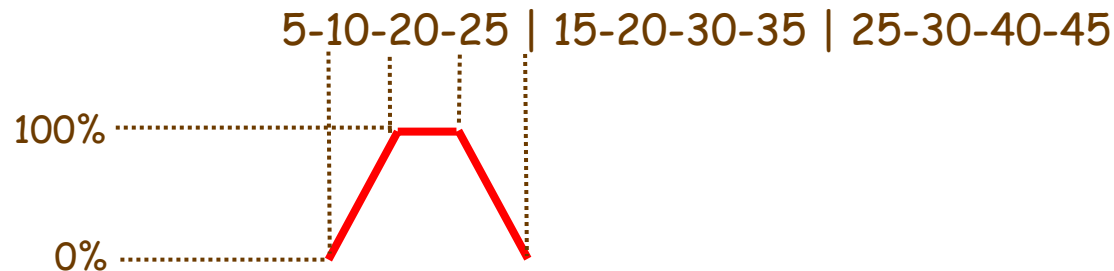
Nyquist: $F_{\max} \approx 1/(2\Delta t)$ or 62 Hz

Exercise: BandPass Filters & Frequency

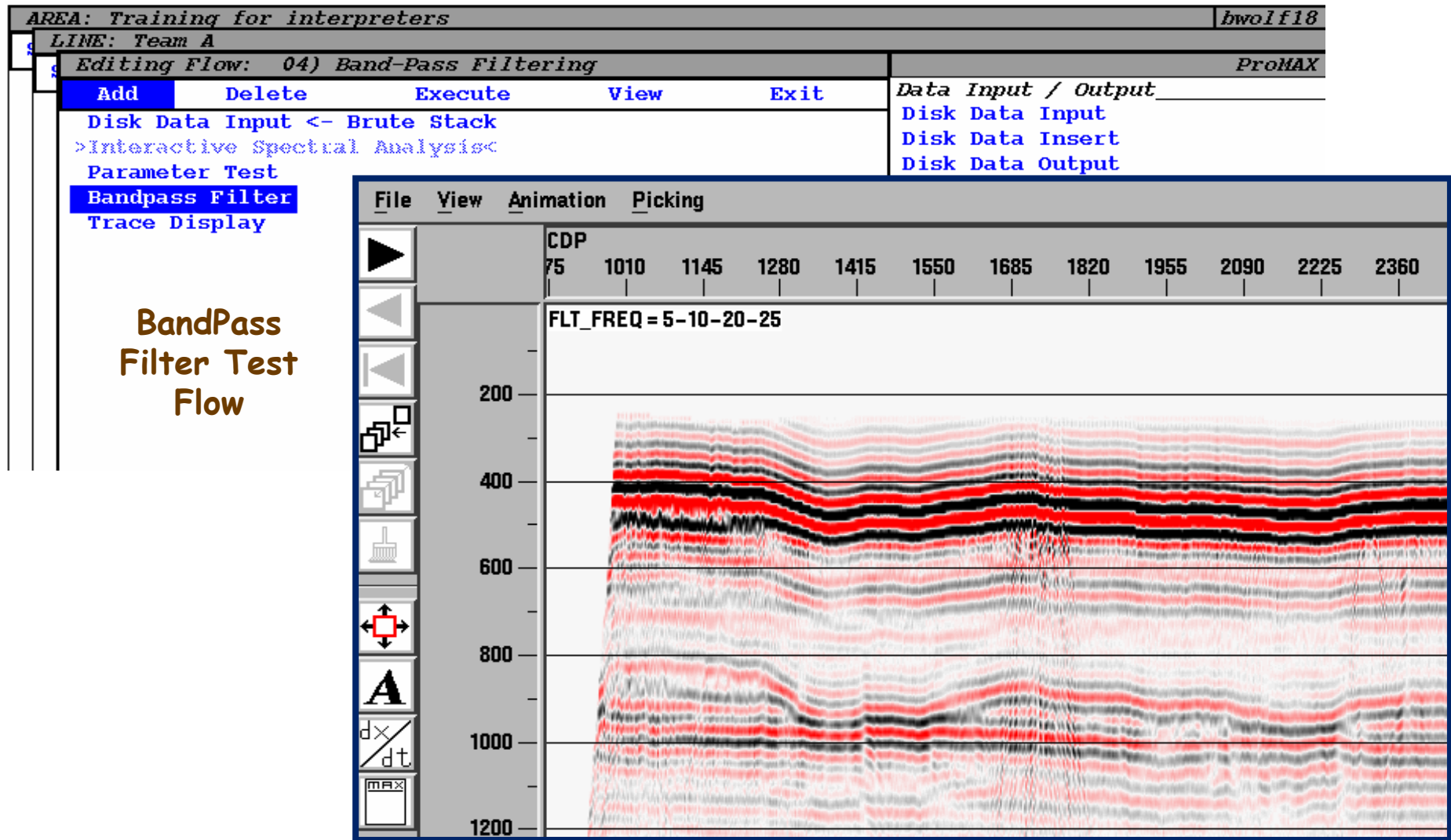
AREA: Training for interpreters		bwolf18																											
LINE: Team A																													
Editing Flow: 04) Band-Pass Filtering		ProMAX																											
Add Delete Execute View Exit Disk Data Input <- Brute Stack >Interactive Spectral Analysis< Parameter Test Bandpass Filter Trace Display	Data Input / Output Disk Data Input Disk Data Insert Disk Data Output SEG II T																												
<table border="1"> <thead> <tr> <th colspan="2">Bandpass Filter</th> <th>?</th> </tr> </thead> <tbody> <tr> <td>TYPE of filter</td> <td colspan="2">Single Filter</td> </tr> <tr> <td>Type of filter specification</td> <td colspan="2">Ormsby bandpass</td> </tr> <tr> <td>PHASE of filter</td> <td colspan="2">Zero</td> </tr> <tr> <td>Domain for filter application</td> <td colspan="2">Frequency</td> </tr> <tr> <td>Percent zero padding for FFT's</td> <td colspan="2">25.</td> </tr> <tr> <td>Apply a notch filter?</td> <td colspan="2">Yes No</td> </tr> <tr> <td>Ormsby filter frequency values</td> <td colspan="2">99999</td> </tr> <tr> <td>Re-apply trace mute after filter?</td> <td colspan="2">Yes No</td> </tr> </tbody> </table>			Bandpass Filter		?	TYPE of filter	Single Filter		Type of filter specification	Ormsby bandpass		PHASE of filter	Zero		Domain for filter application	Frequency		Percent zero padding for FFT's	25.		Apply a notch filter?	Yes No		Ormsby filter frequency values	99999		Re-apply trace mute after filter?	Yes No	
Bandpass Filter		?																											
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Percent zero padding for FFT's	25.																												
Apply a notch filter?	Yes No																												
Ormsby filter frequency values	99999																												
Re-apply trace mute after filter?	Yes No																												
Null Data File																													

'99999' refers to the band-pass tests in the *Parameter Test* process

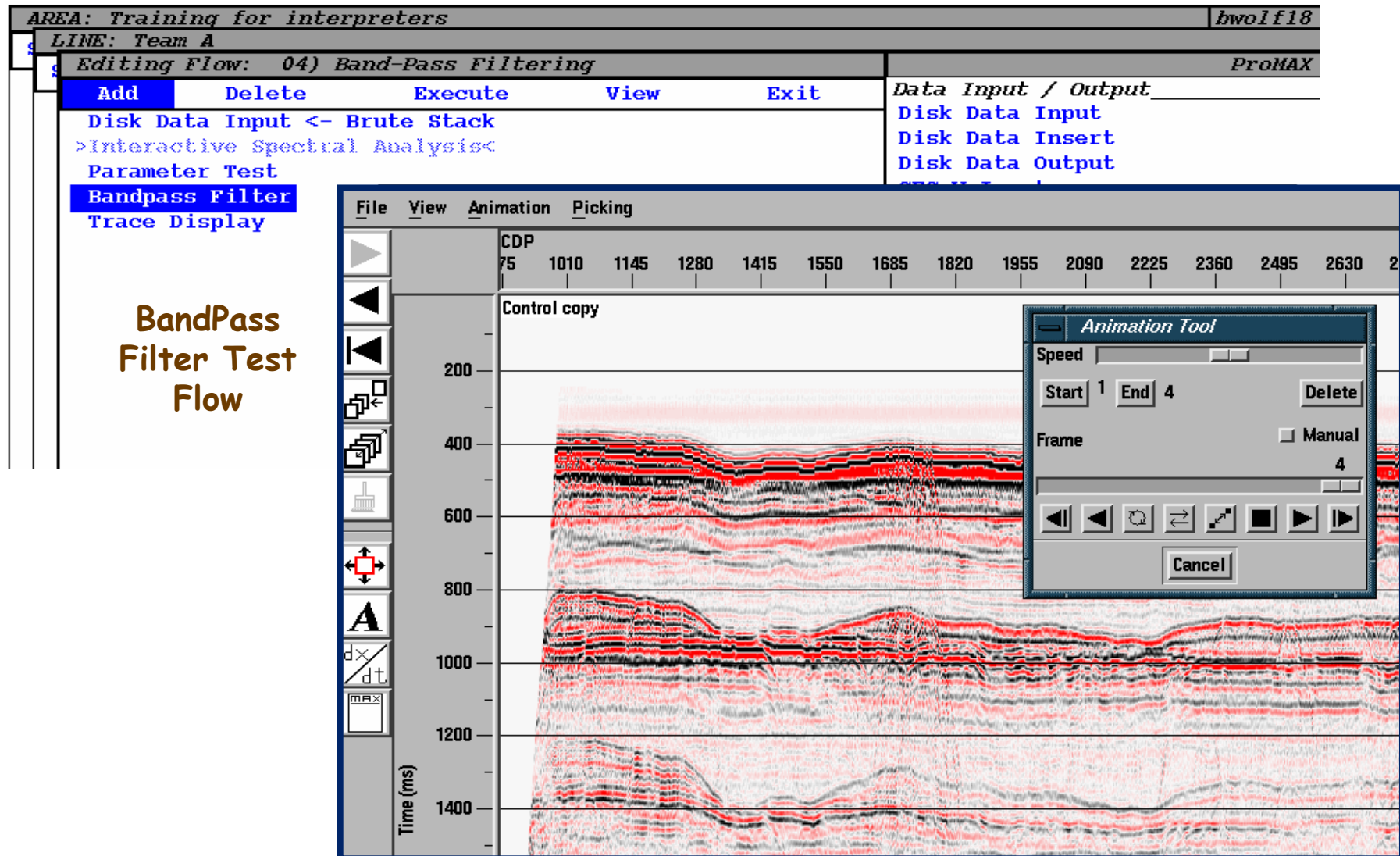
Band-pass test ranges (in Hz):



Exercise: BandPass Filters & Frequency



Exercise: BandPass Filters & Frequency



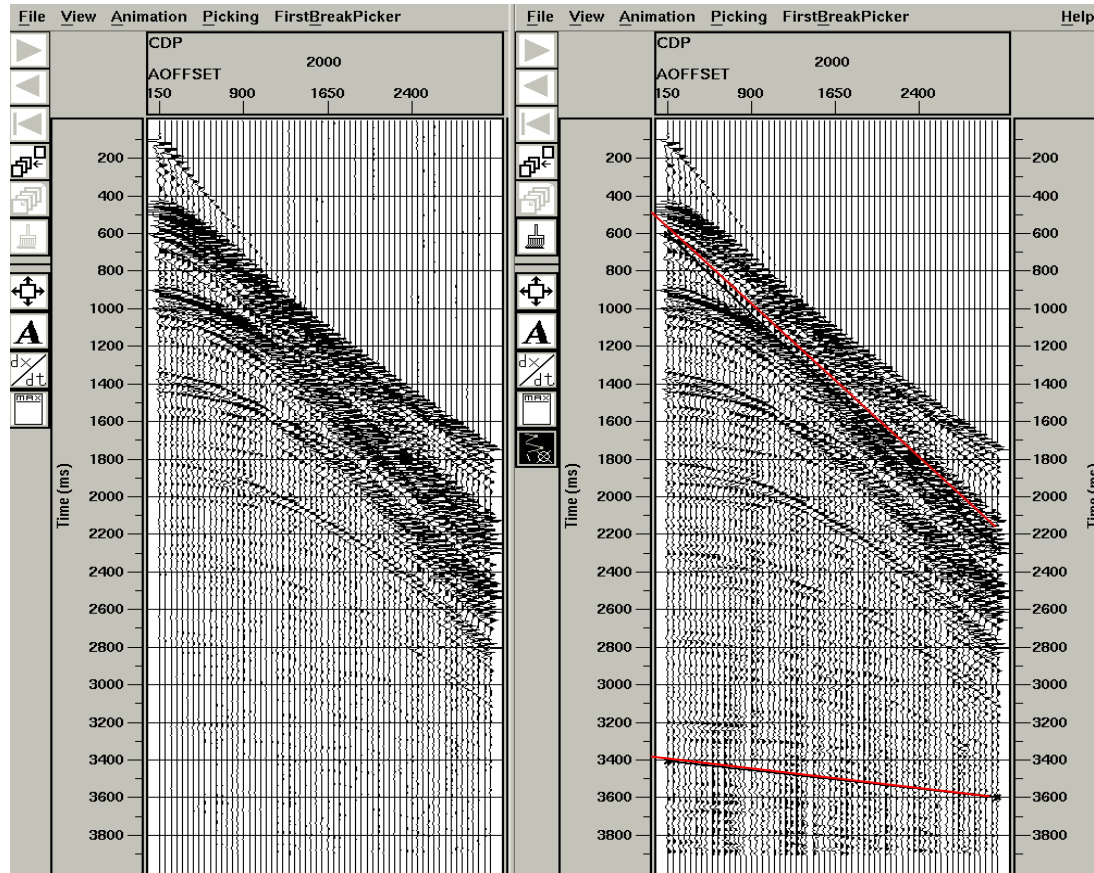
Introduction to exercises:

Deconvolution

Course in

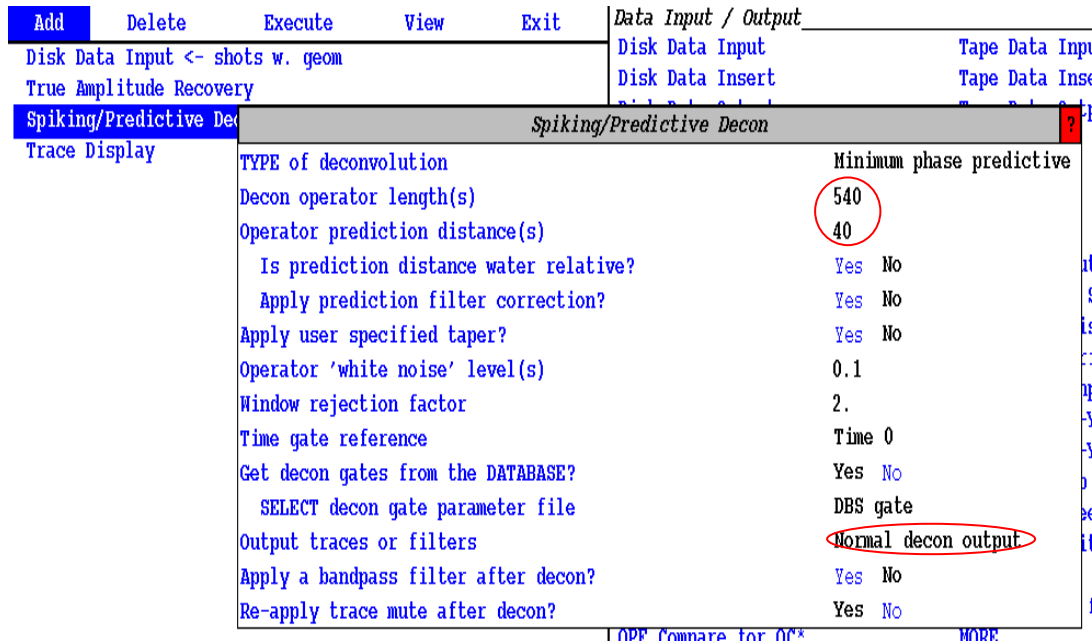
Seismic processing for interpreters

Exercise: Deconvolution



- 1) Apply data independent gain correction to shot gathers
- 2) Pick an autocorrelation window, start time is usually below the strong WB reflector, the window length should include several periods of multiple ringing

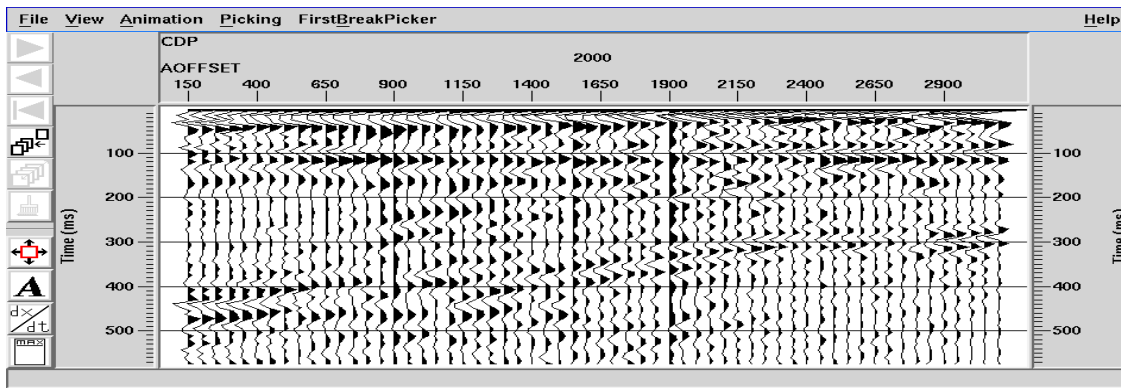
Exercise: Deconvolution



-If you change the output from *Normal decon* output to *Autocorrelations* output you will display the autocorrelation function, shown below to the left

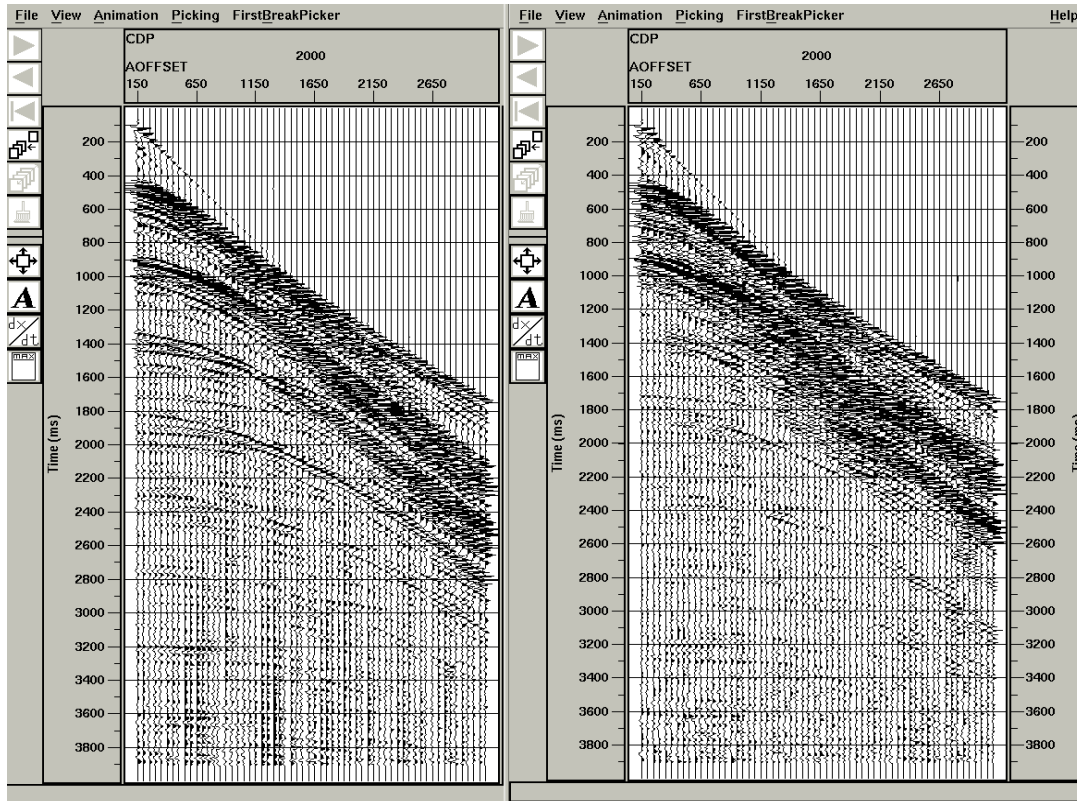
-The *Decon operator length* should be long enough to include the multiple period

-The *Operator prediction distance* (gap) is typically chosen to be as long as the second autocorrelation function zero crossing

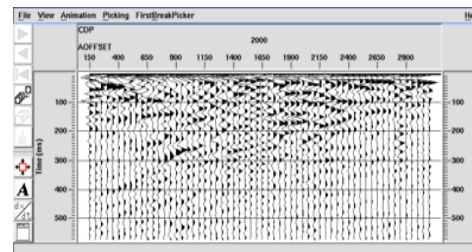
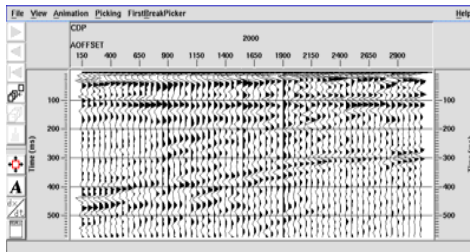


Autocorrelation functions

Exercise: Deconvolution

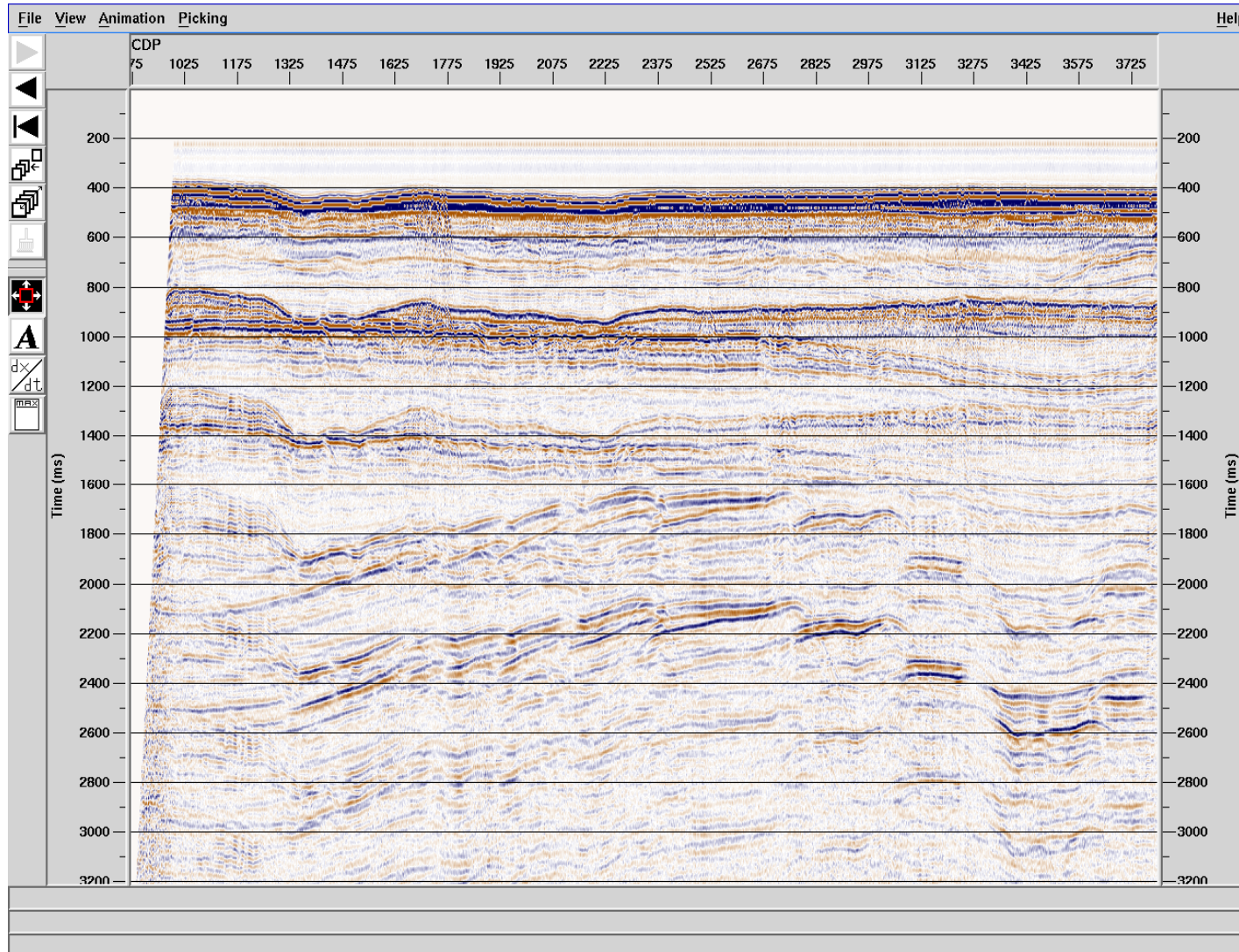


- Compare a shot gather before (left) and after (right) the application of predictive decon
- Autocorrelation functions are shown below



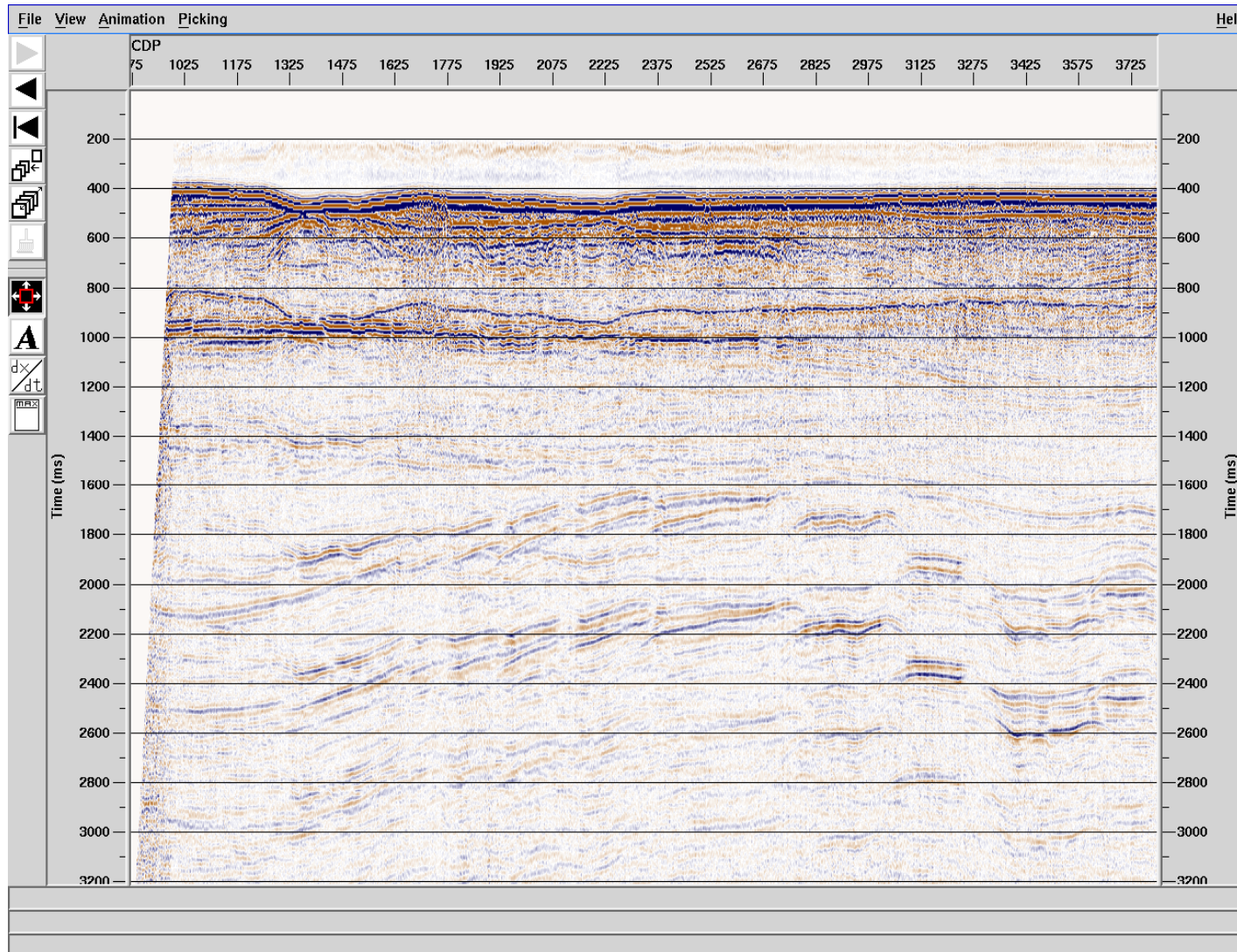
Autocorrelation functions

Exercise: Deconvolution



-Brute Stack

Exercise: Deconvolution



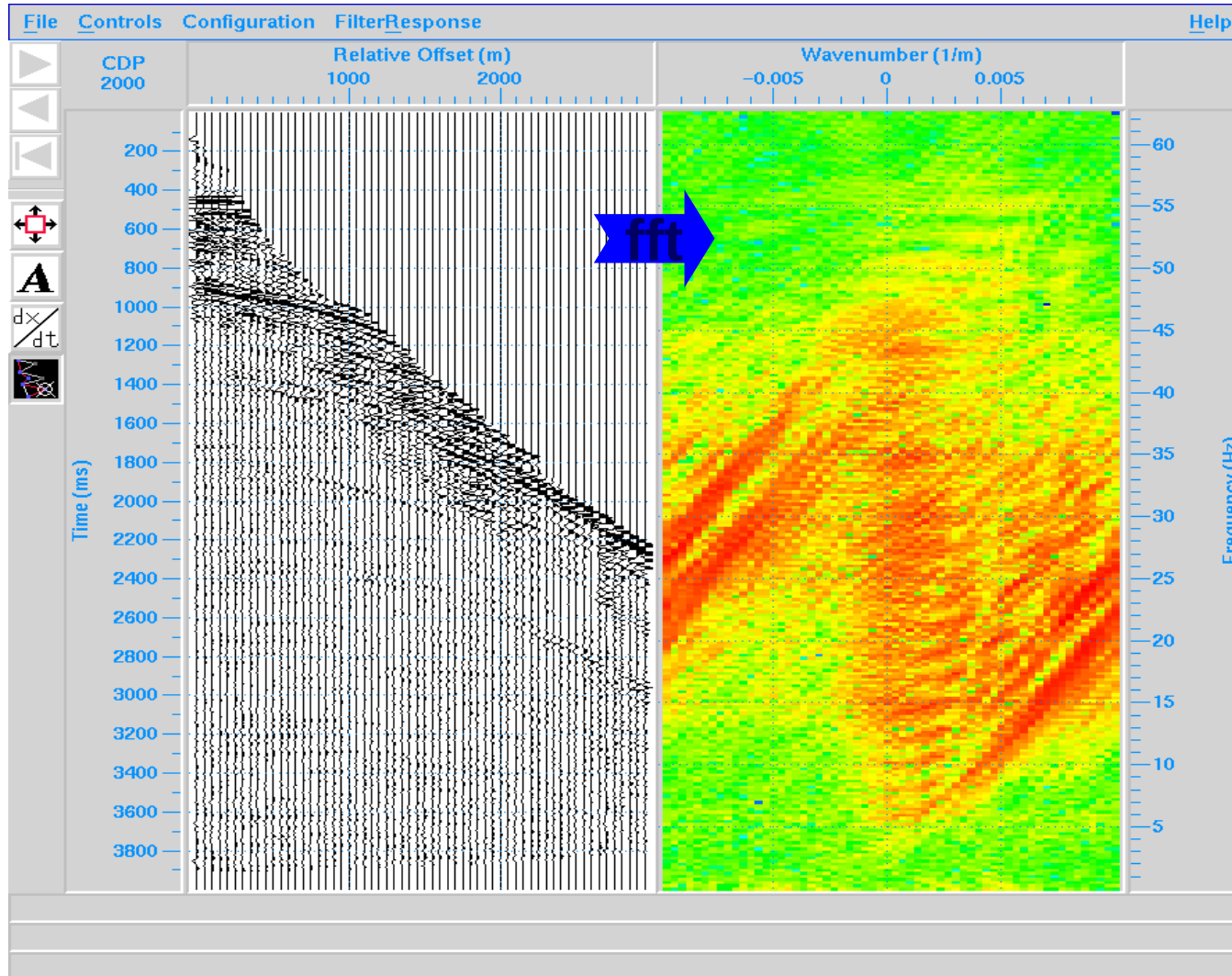
-Brute Stack w/DBS
(Deconvolution Before Stack)

Introduction to exercises:

FK Filtering

Course in
Seismic processing for interpreters

Exercise: FK filtering

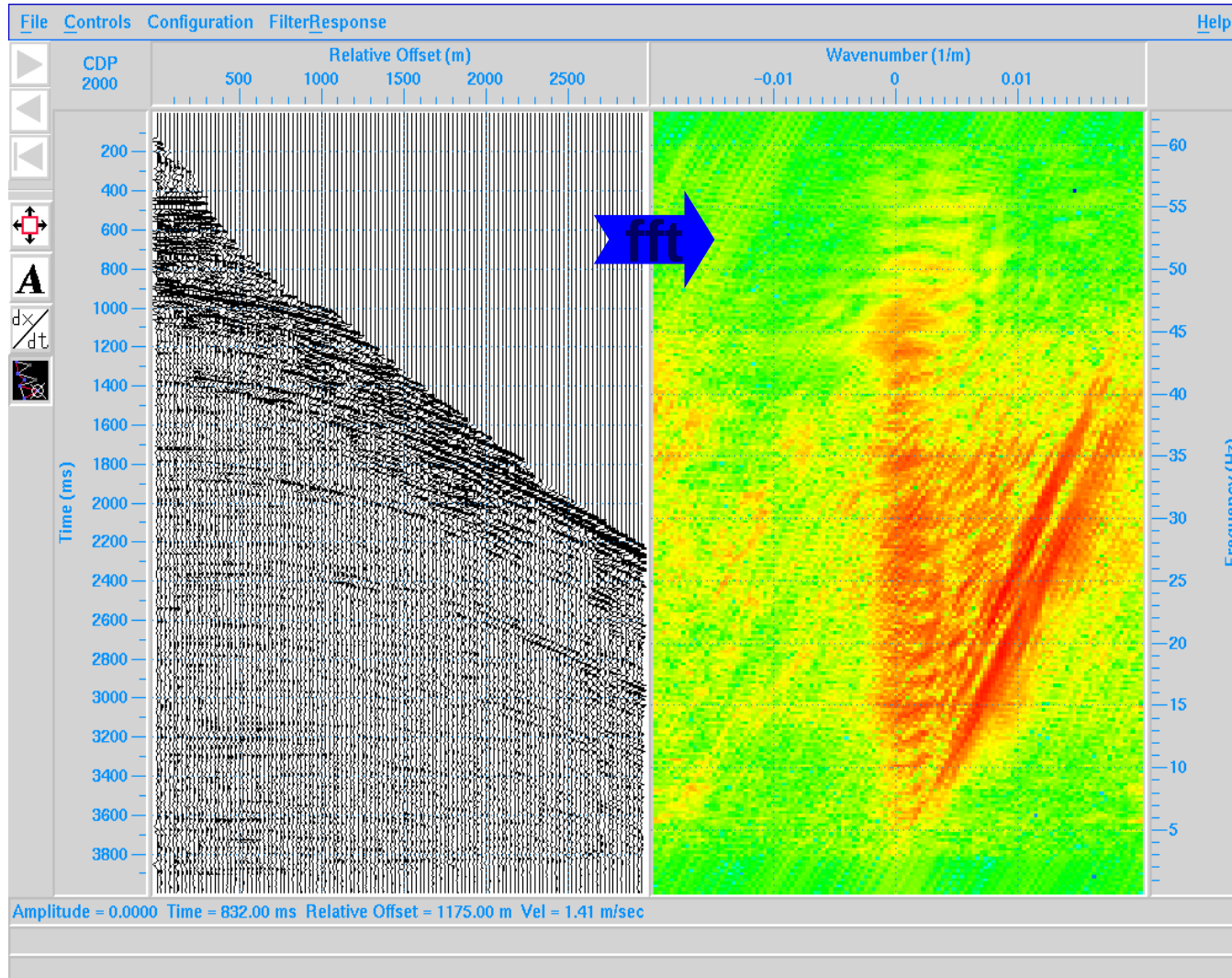


1. NMO correct a CDP gather

2. Apply FK analysis

Note strong evidence of spatial aliasing in the FK domain

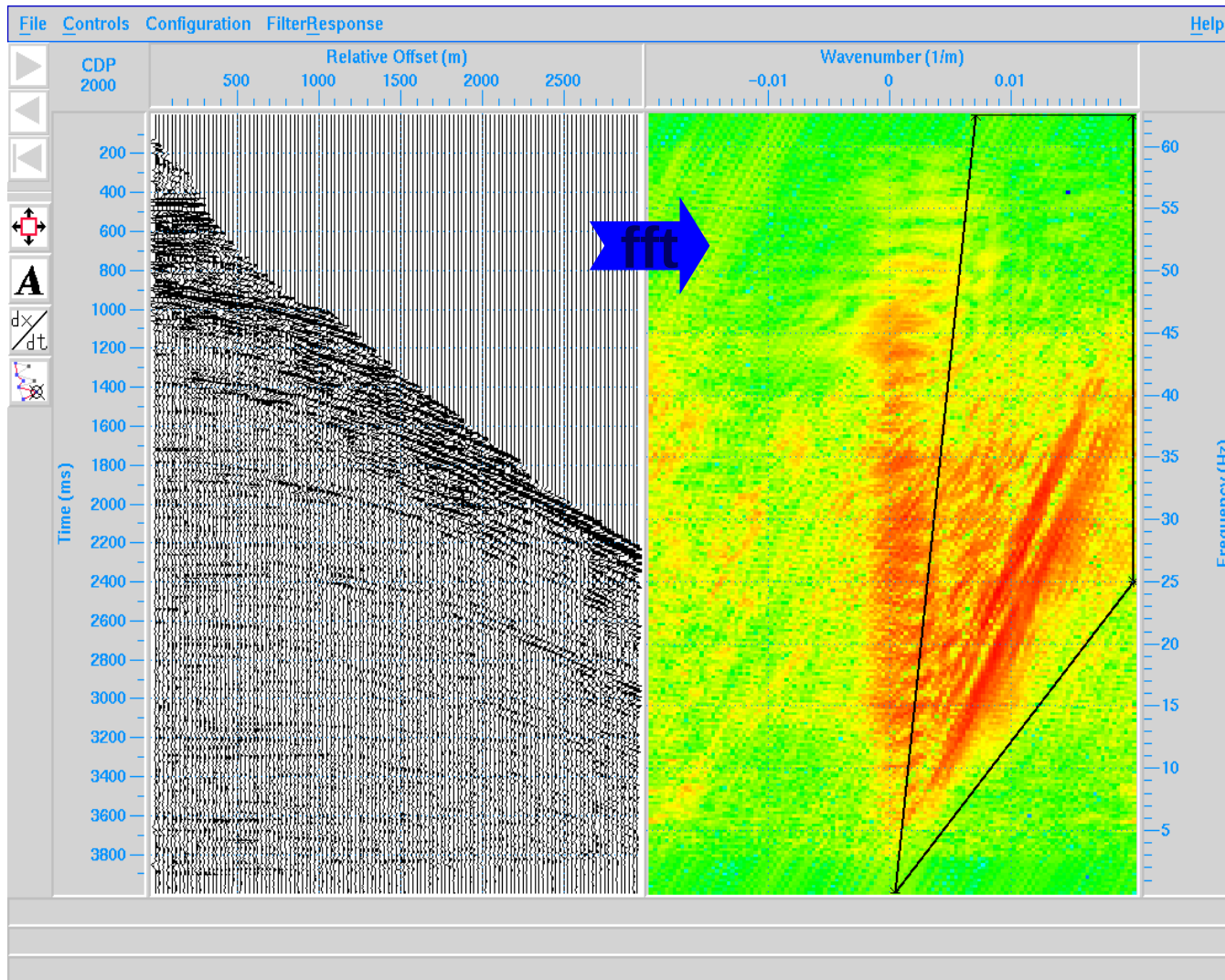
Exercise: FK filtering



1. Go back and do dip dependent FX interpolation from 50m trace spacing to 25m trace spacing
2. Apply FK analysis (remember to change the trace spacing and the panel width)

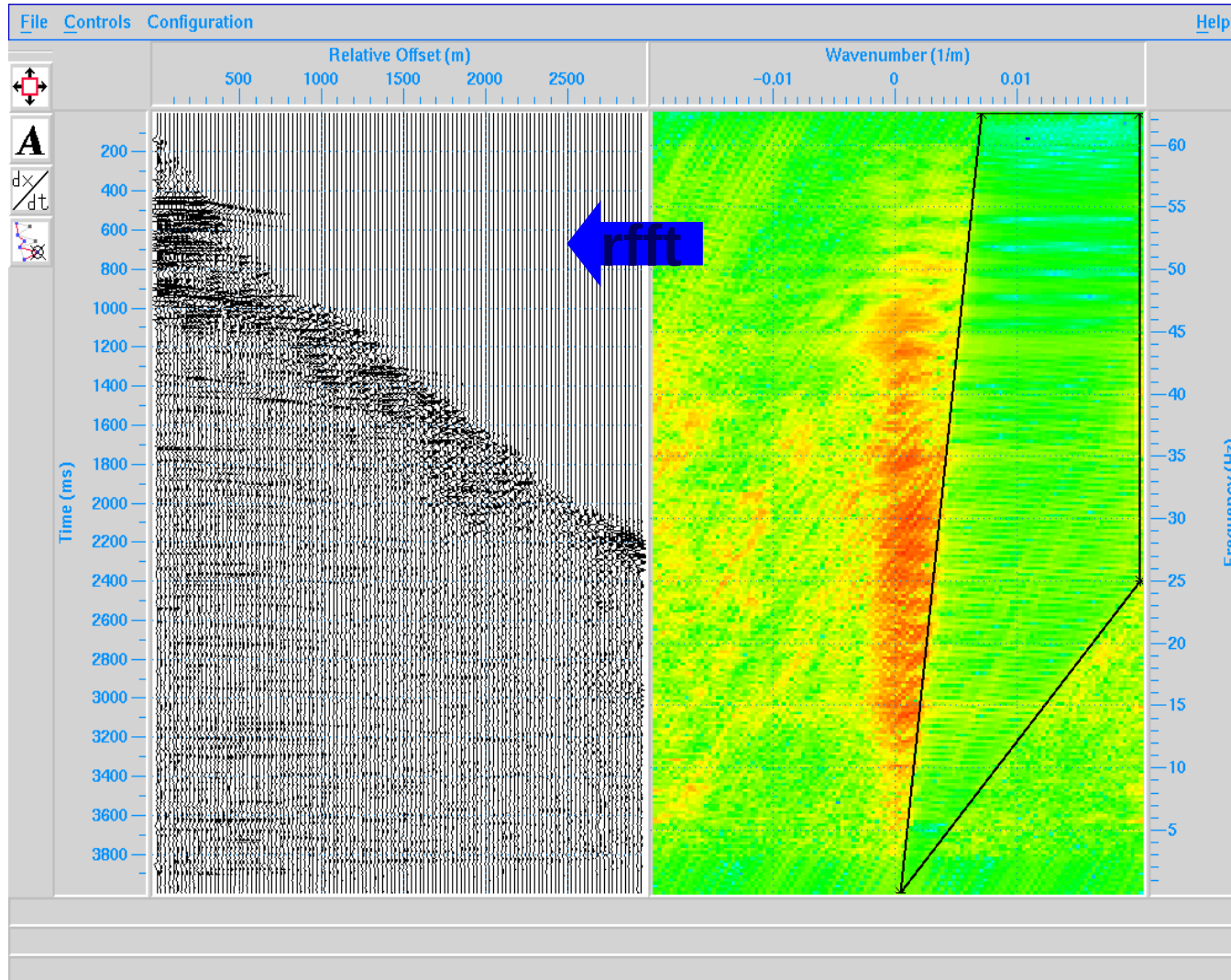
Note how spatial aliasing has been reduced

Exercise: FK filtering



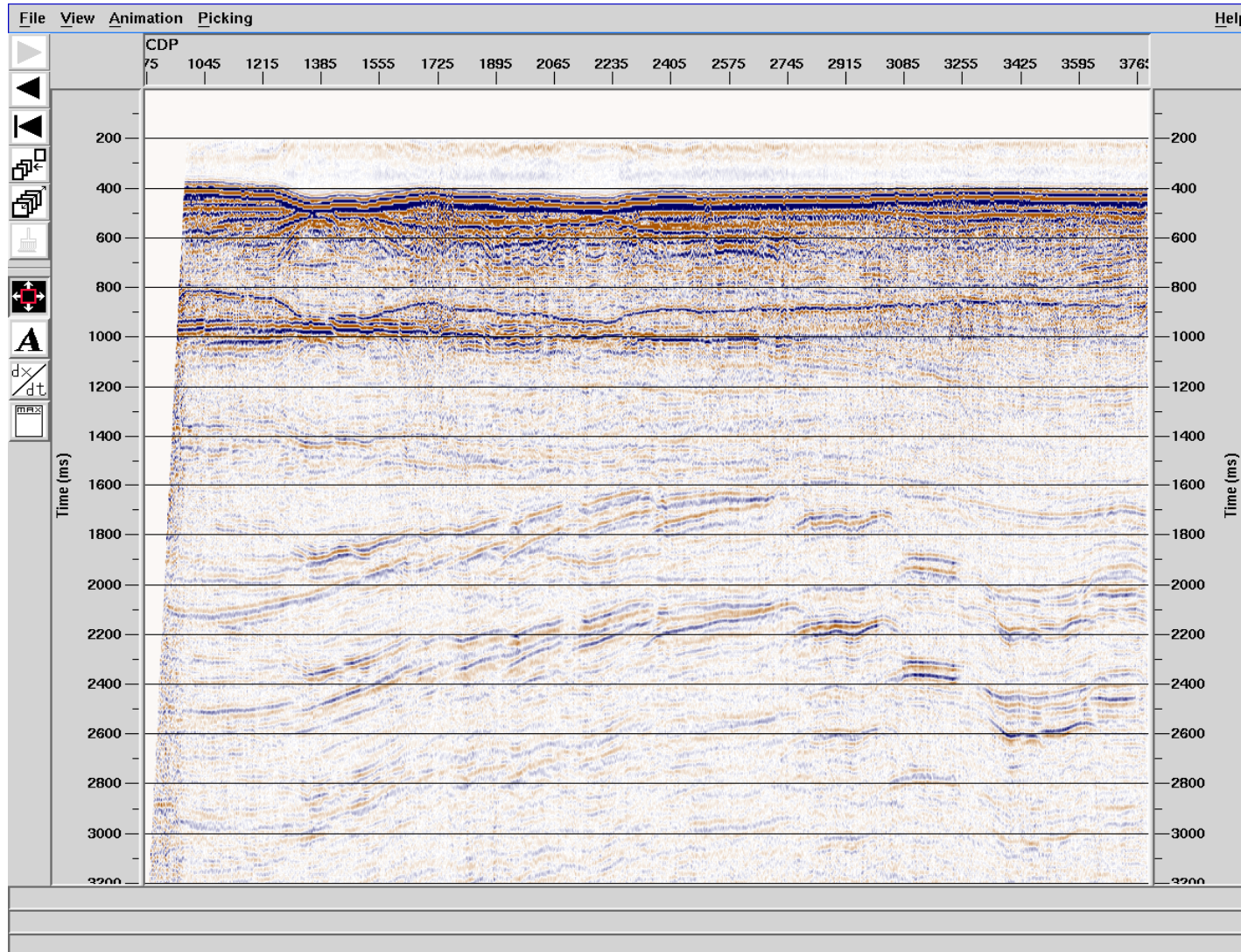
Pick a reject fan (a pie slice) including the multiple energy

Exercise: FK filtering



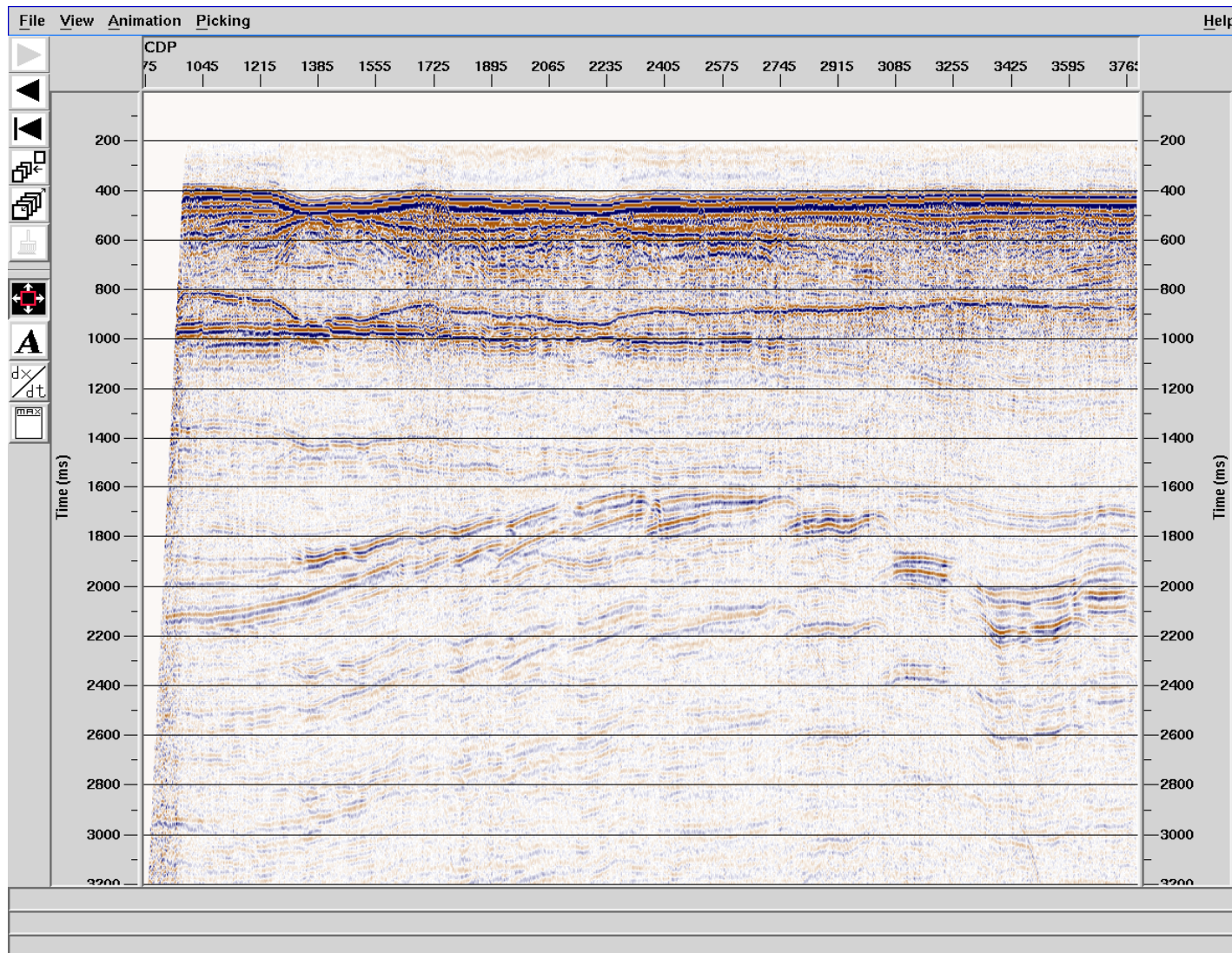
1. Apply the FK filter
 2. Do a reverse Fourier transform
- Observe how the multiples are attenuated

Exercise: FK filtering



Brute Stack w/DBS
(Deconvolution Before Stack)

Exercise: FK filtering



Brute Stack w/DBS
(Deconvolution Before Stack)
and FK-demultiple

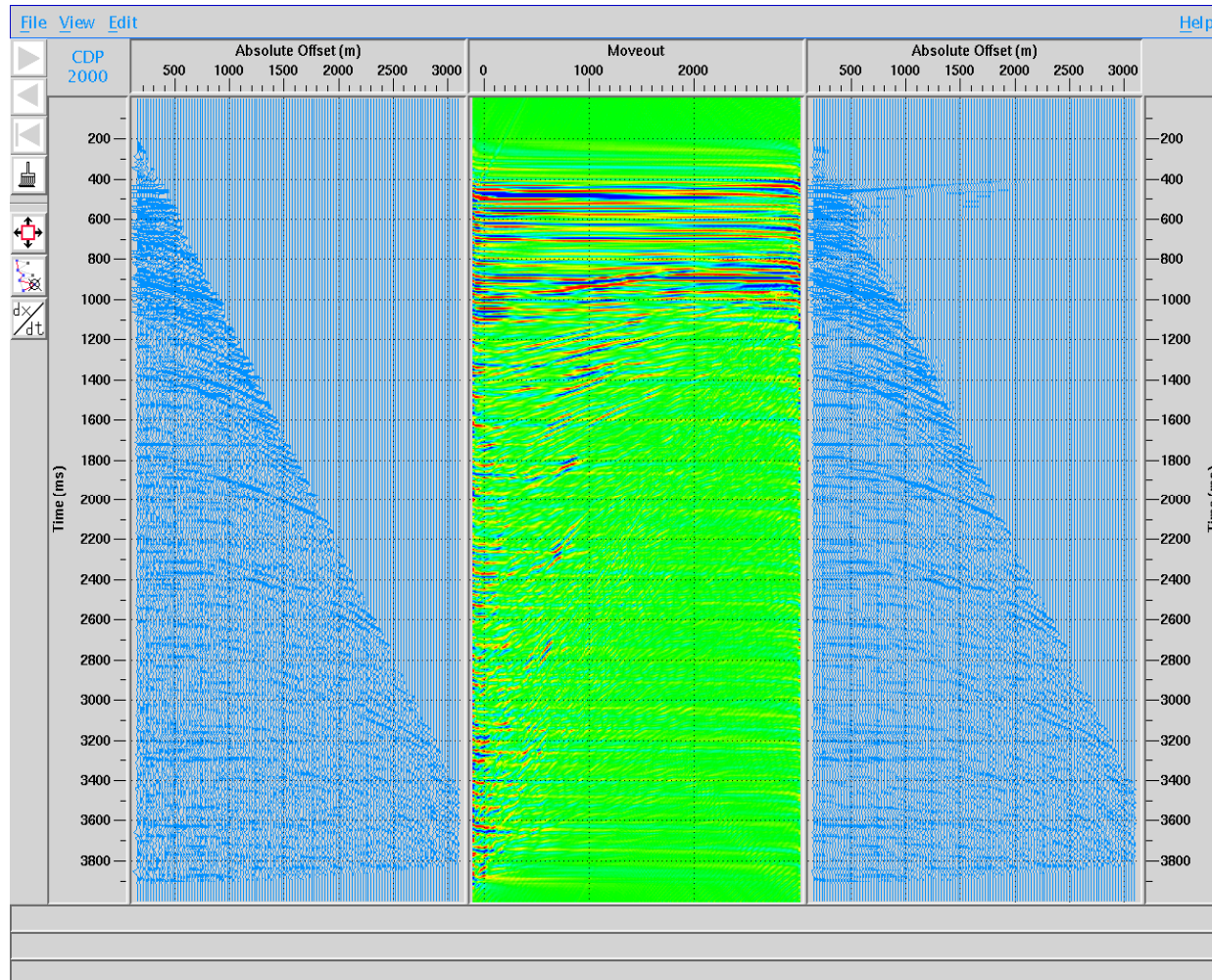
Introduction to exercises:

Radon Filtering

Course in

Seismic processing for interpreters

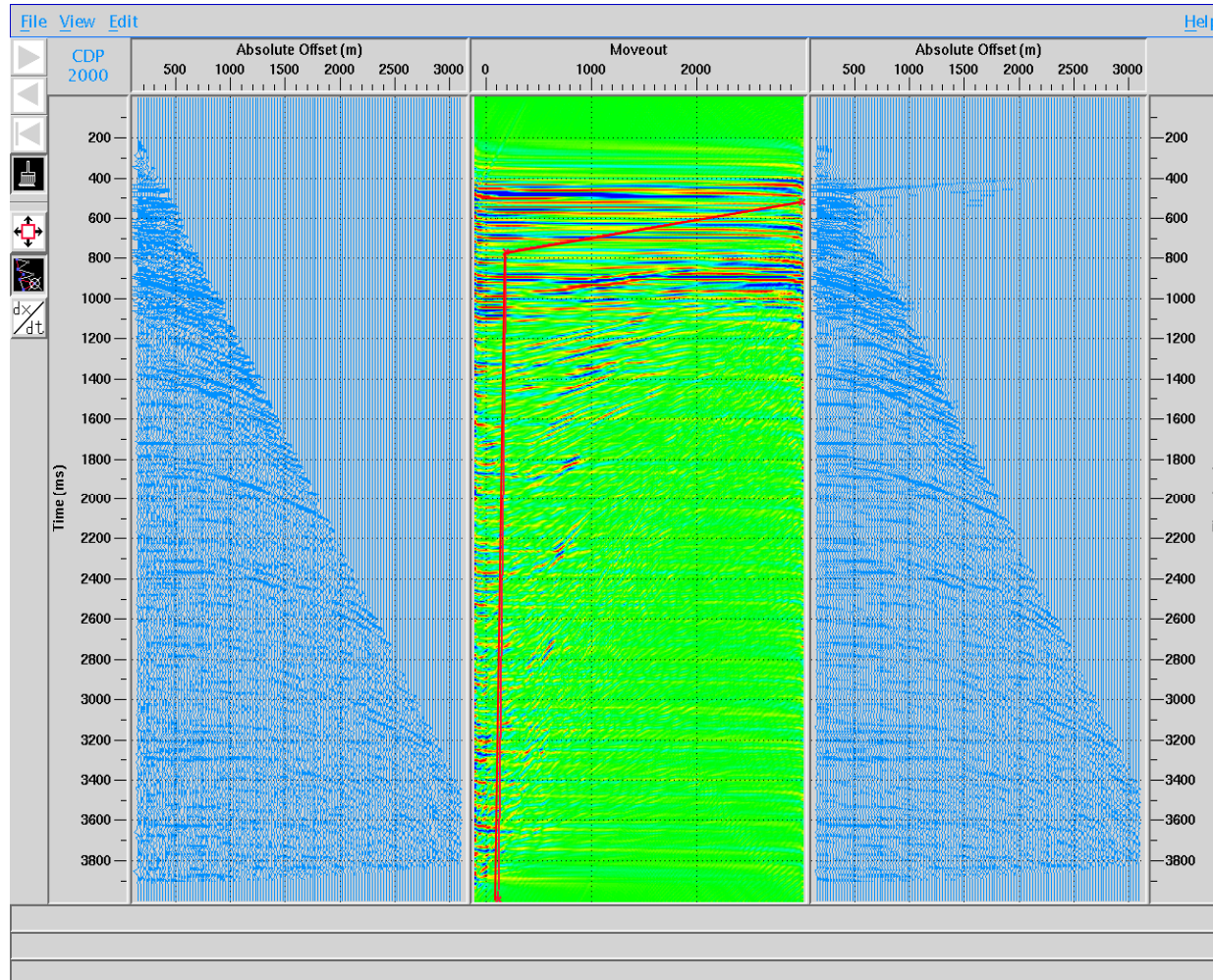
Exercise: Parabolic radon filtering



1. NMO correct a CDP gather and interpolate
2. Run/apply Radon Analysis

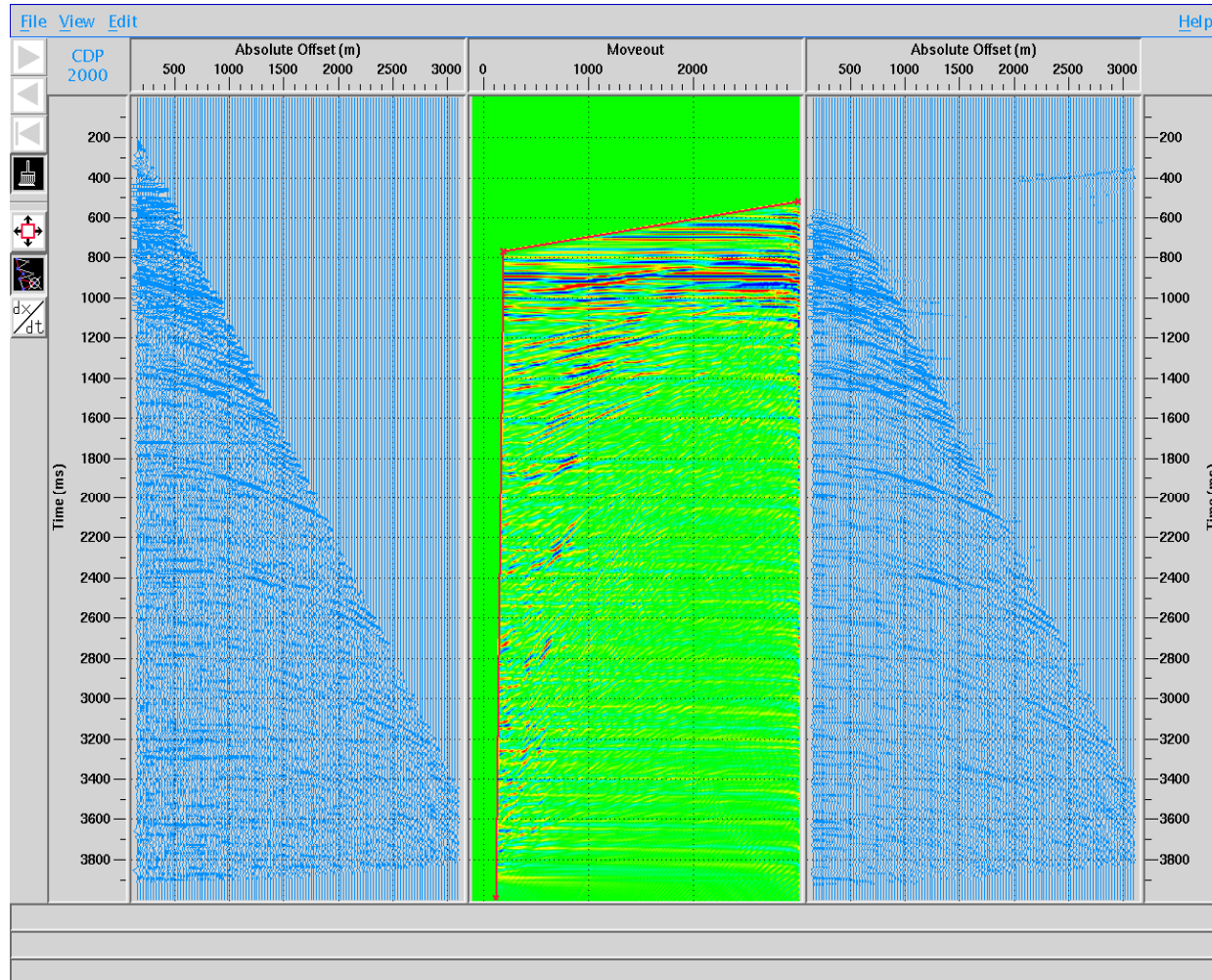
The panels show input CDP gather (l), forward radon transform (m), forward and reverse radon transform (r)

Exercise: Parabolic radon filtering



Pick a mute in the radon domain separating primaries from multiples

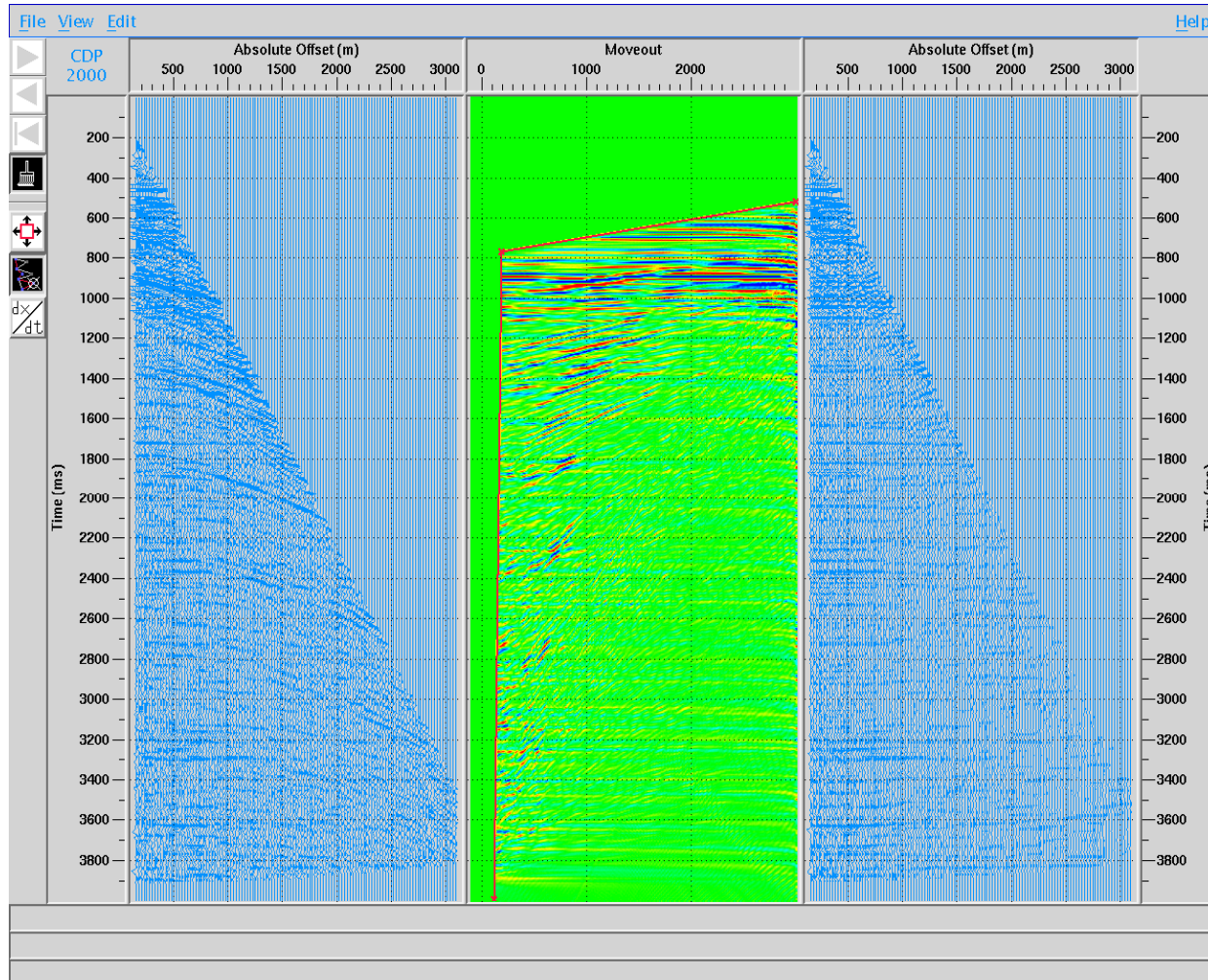
Exercise: Parabolic radon filtering



Remove the primaries

The right panel shows the multiple model

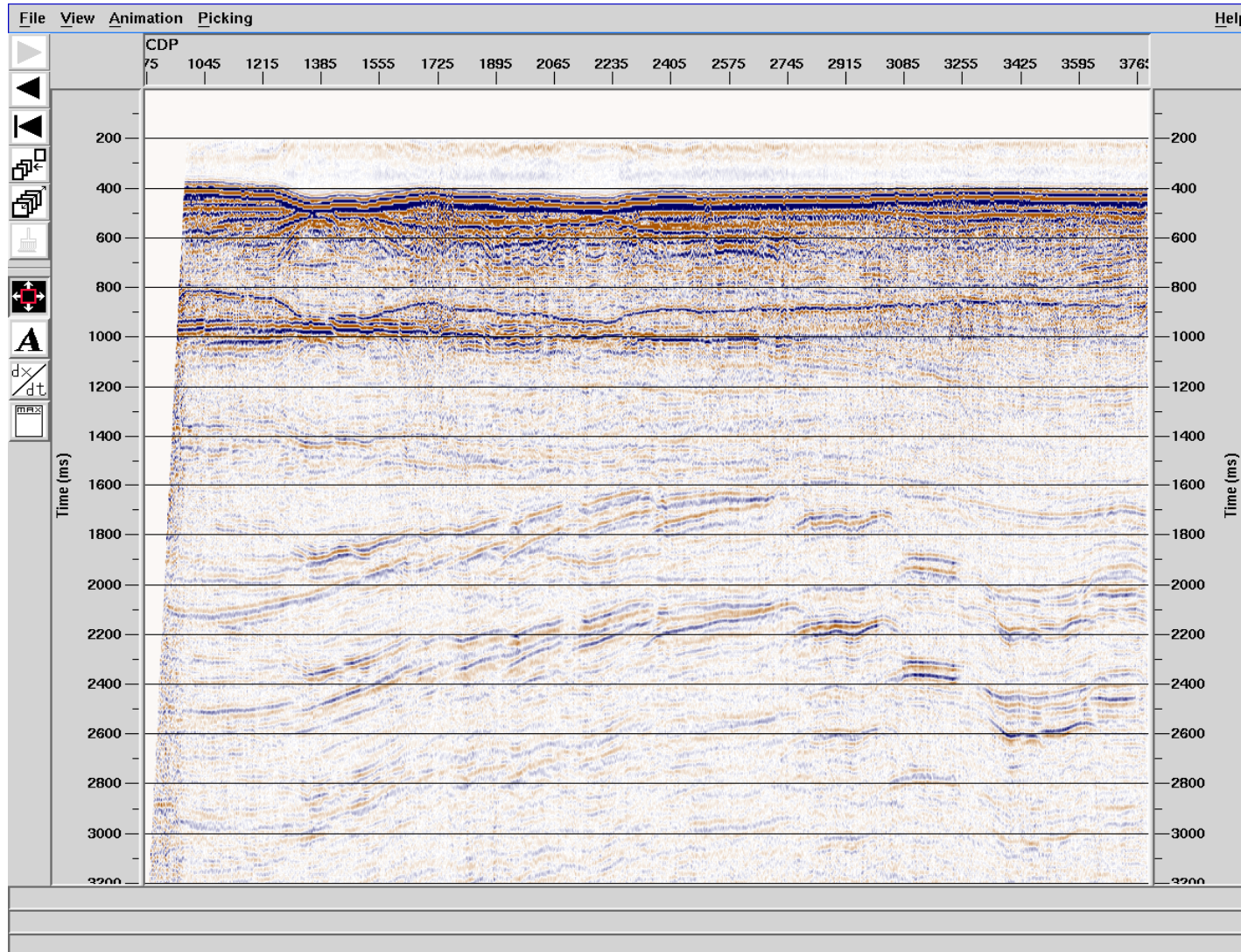
Exercise: Parabolic radon filtering



Remove the primaries

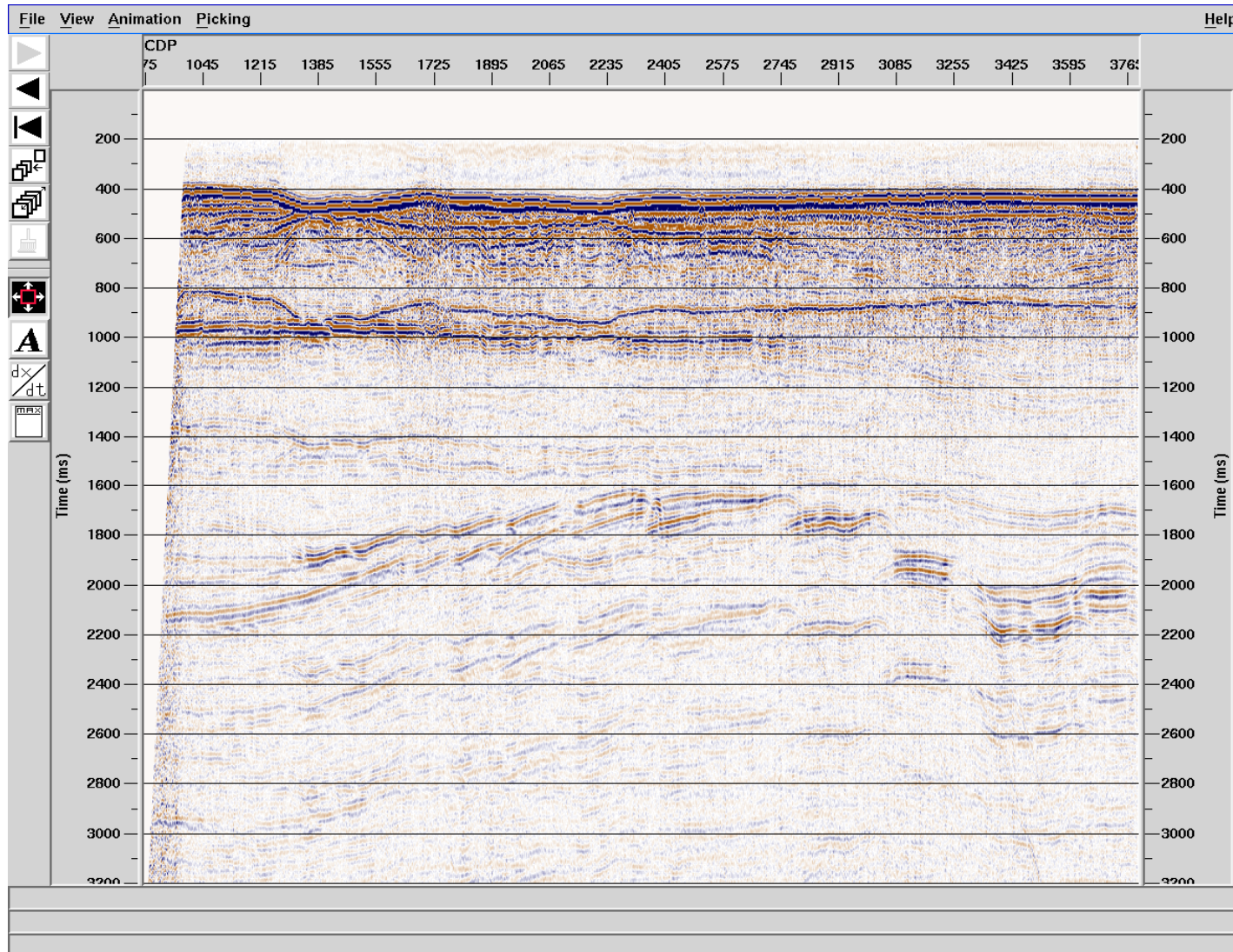
The right panel shows the multiple model subtracted from the input CDP gather

Exercise: Parabolic radon filtering



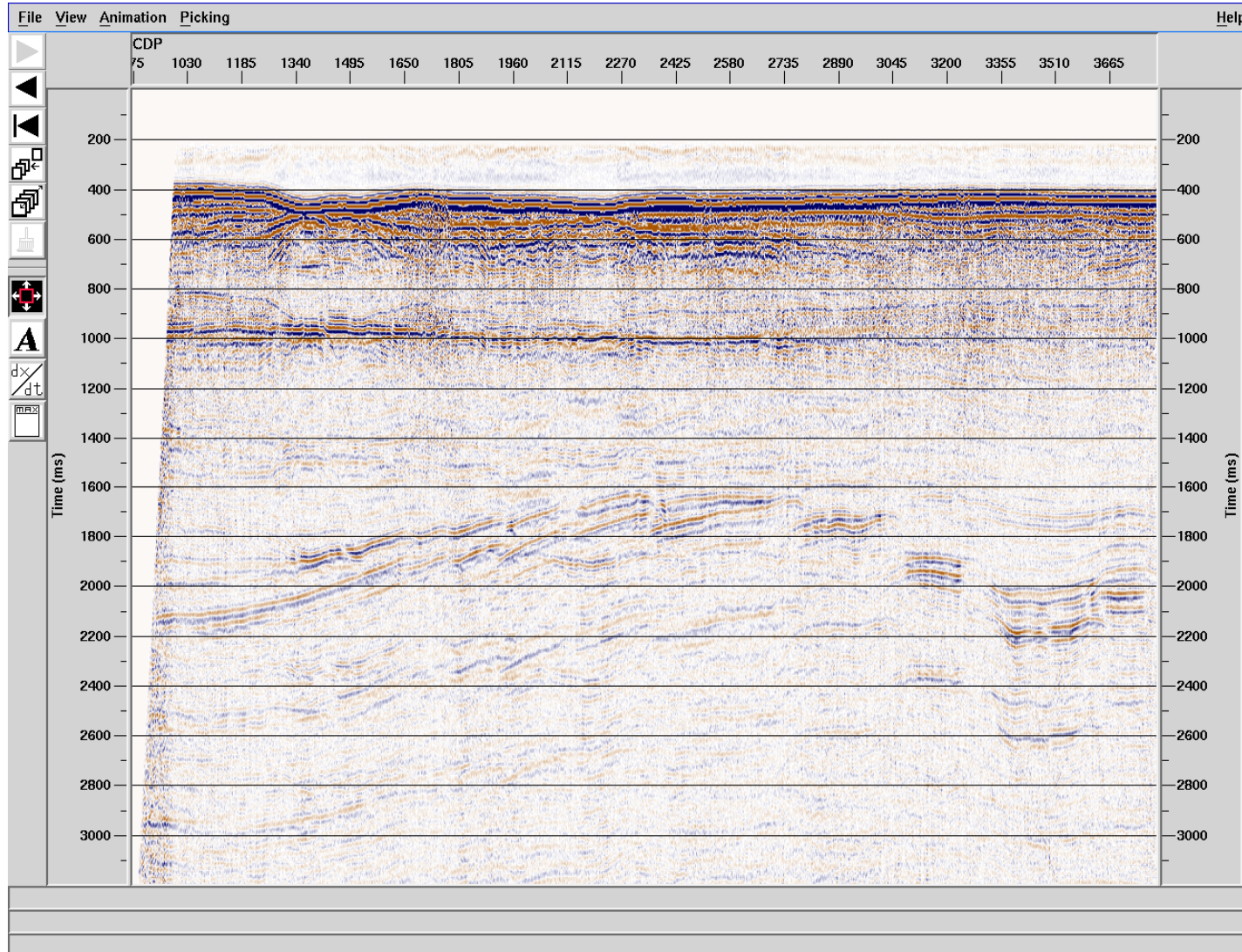
Brute Stack w/DBS
(Deconvolution Before Stack)

Exercise: Parabolic radon filtering



Brute Stack w/DBS
(Deconvolution Before Stack)
+ FK demultiple

Exercise: Parabolic radon filtering



Brute Stack w/DBS
(Deconvolution Before Stack)
+ Radon demultiple

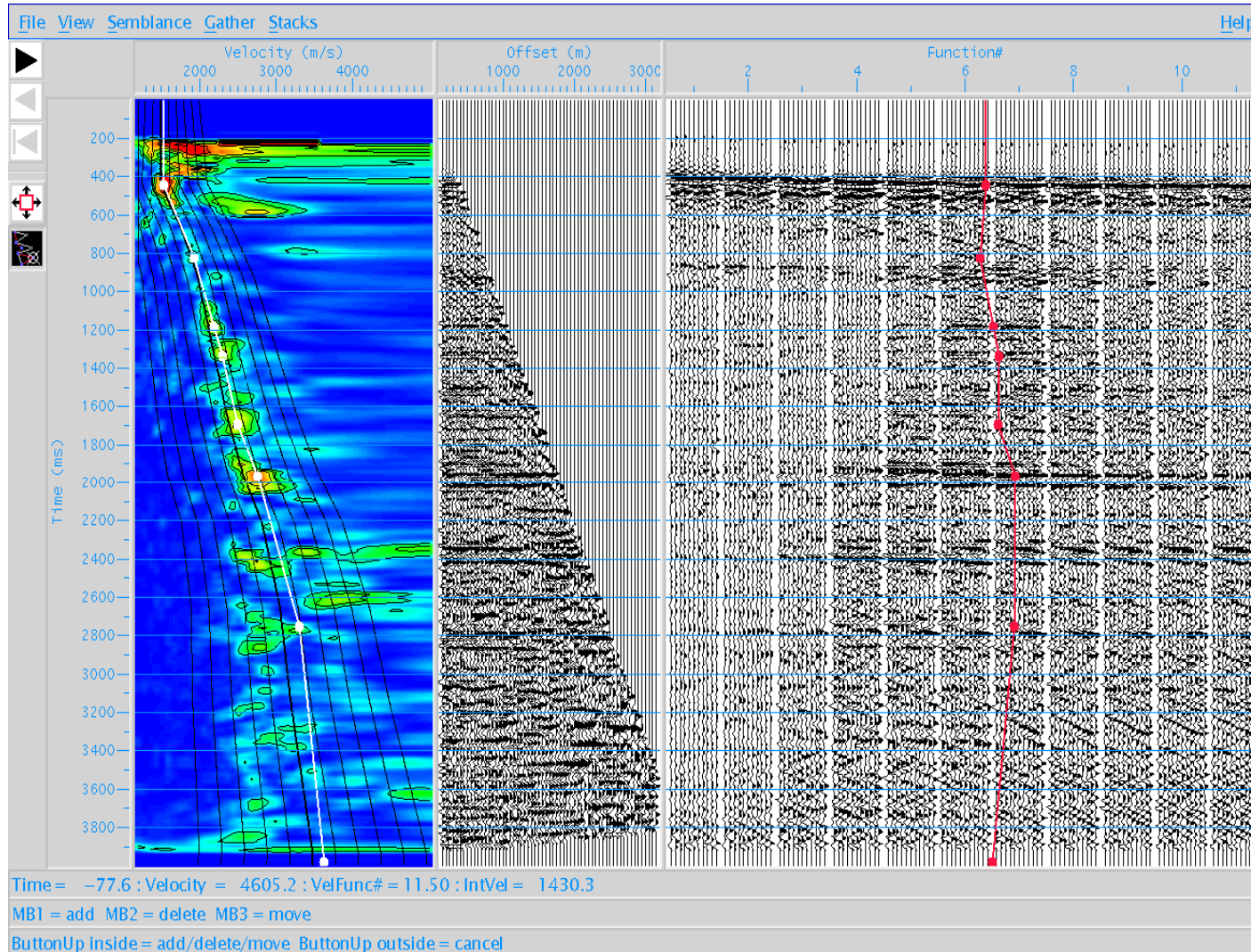
Introduction to exercises:

Velocity Analysis

Course in

Seismic processing for interpreters

Exercise: Velocity analysis



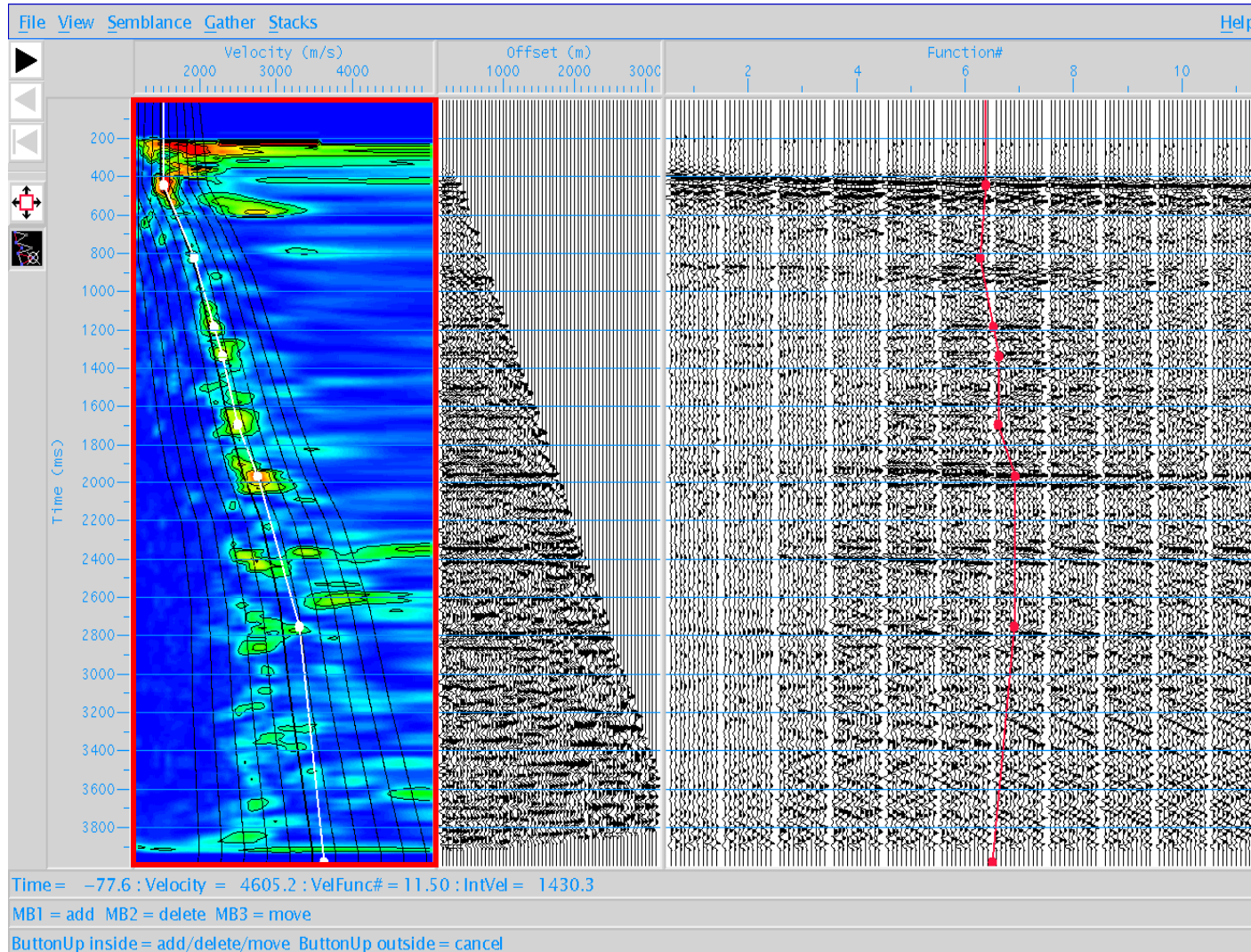
Create supergathers:

3 summed gathers are used as input to semblance calculation

11 gathers are used to form velocity function stack panel

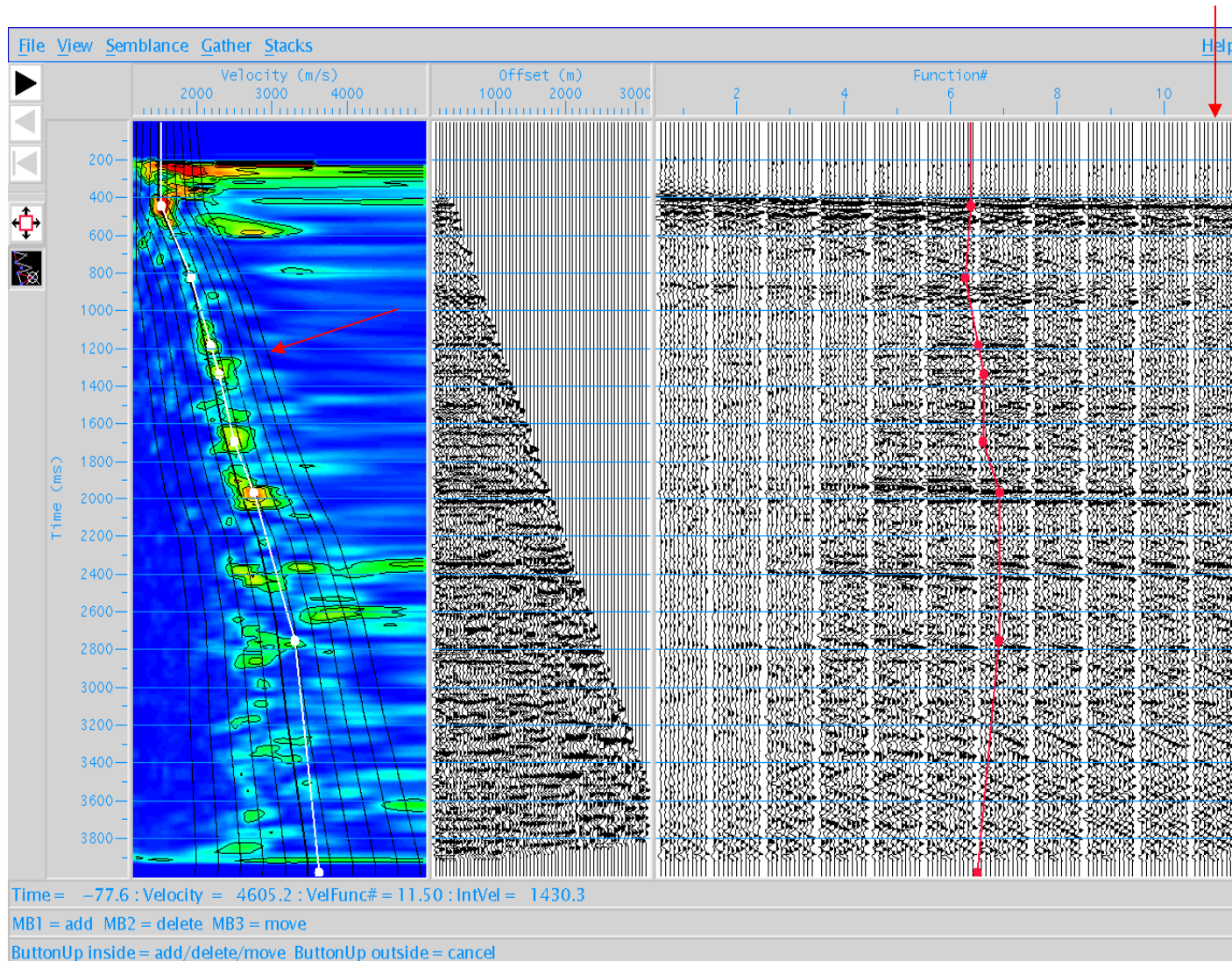
Apply bandpass filter & AGC

Exercise: Velocity analysis



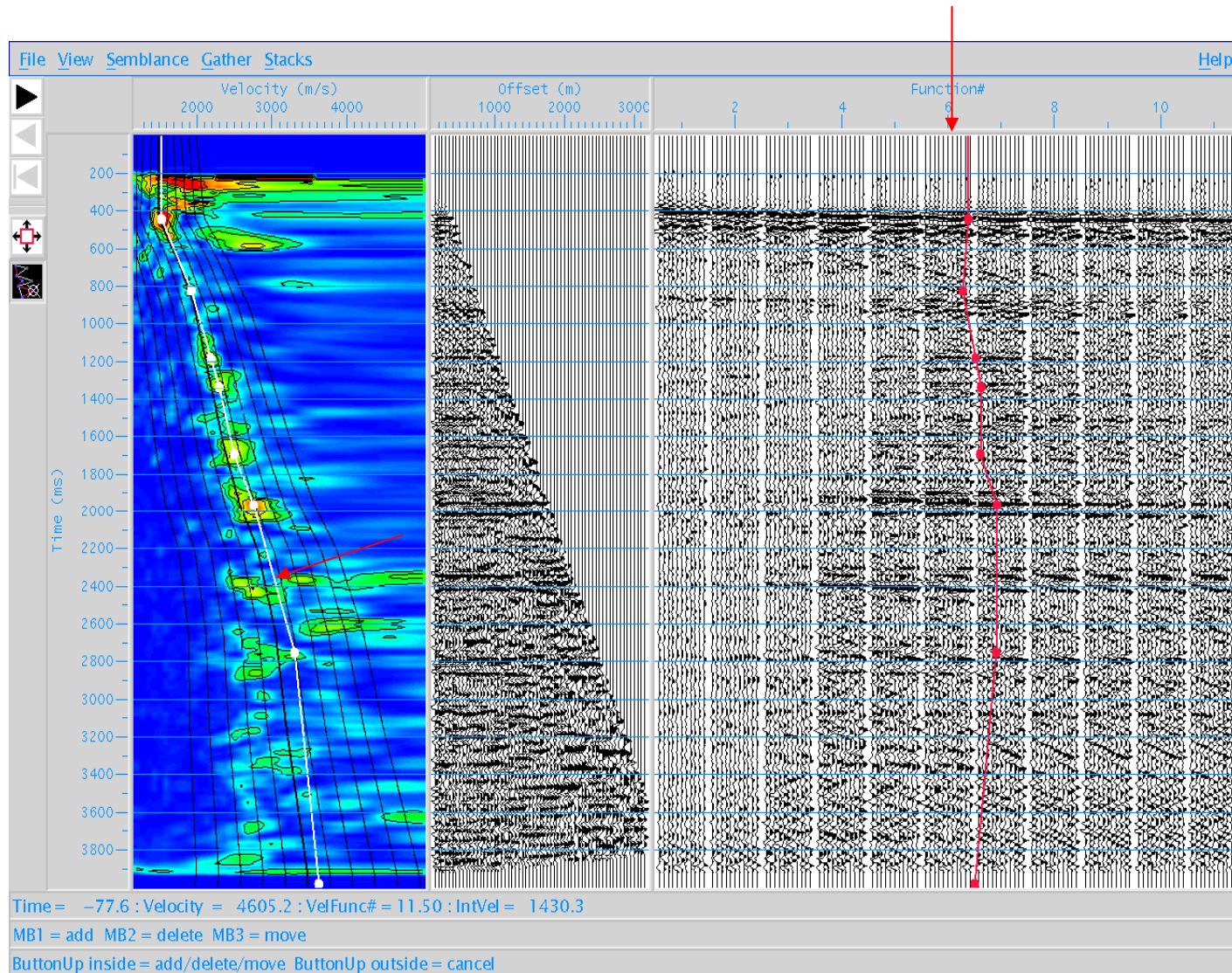
Semblance panel

Exercise: Velocity analysis



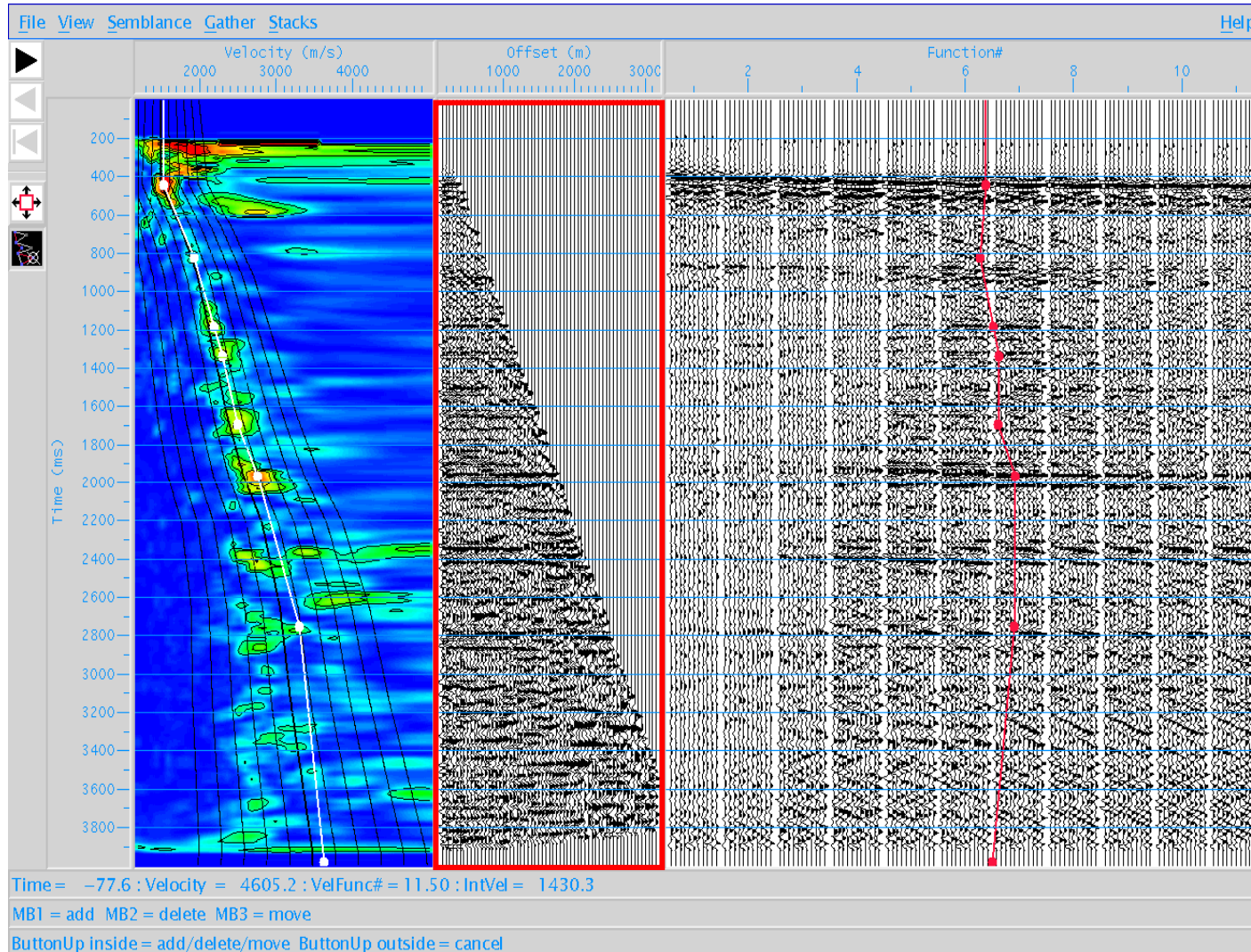
Stack velocity functions

Exercise: Velocity analysis



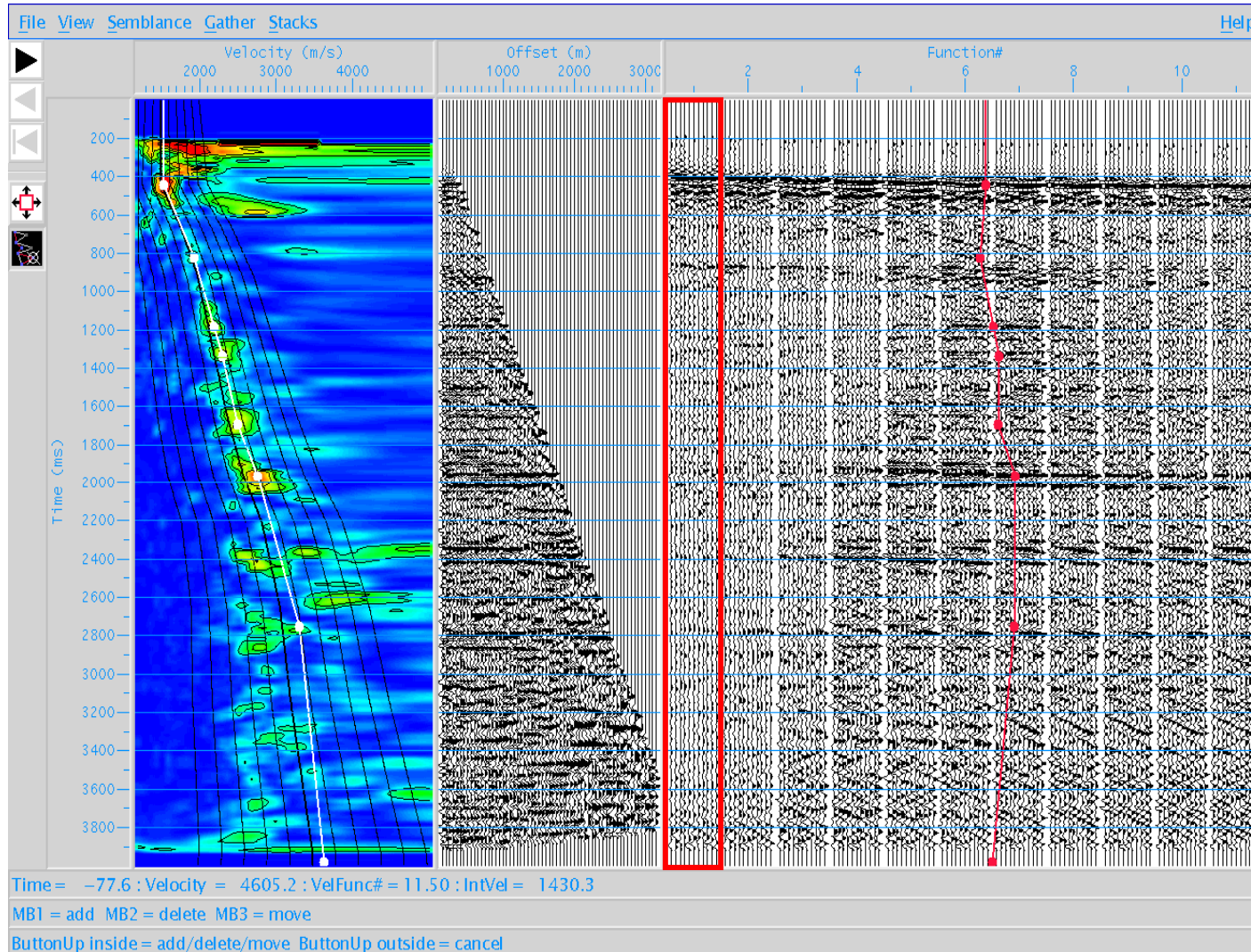
Velocity guide function

Exercise: Velocity analysis



Gather panel,
interactively NMO
corrected

Exercise: Velocity analysis



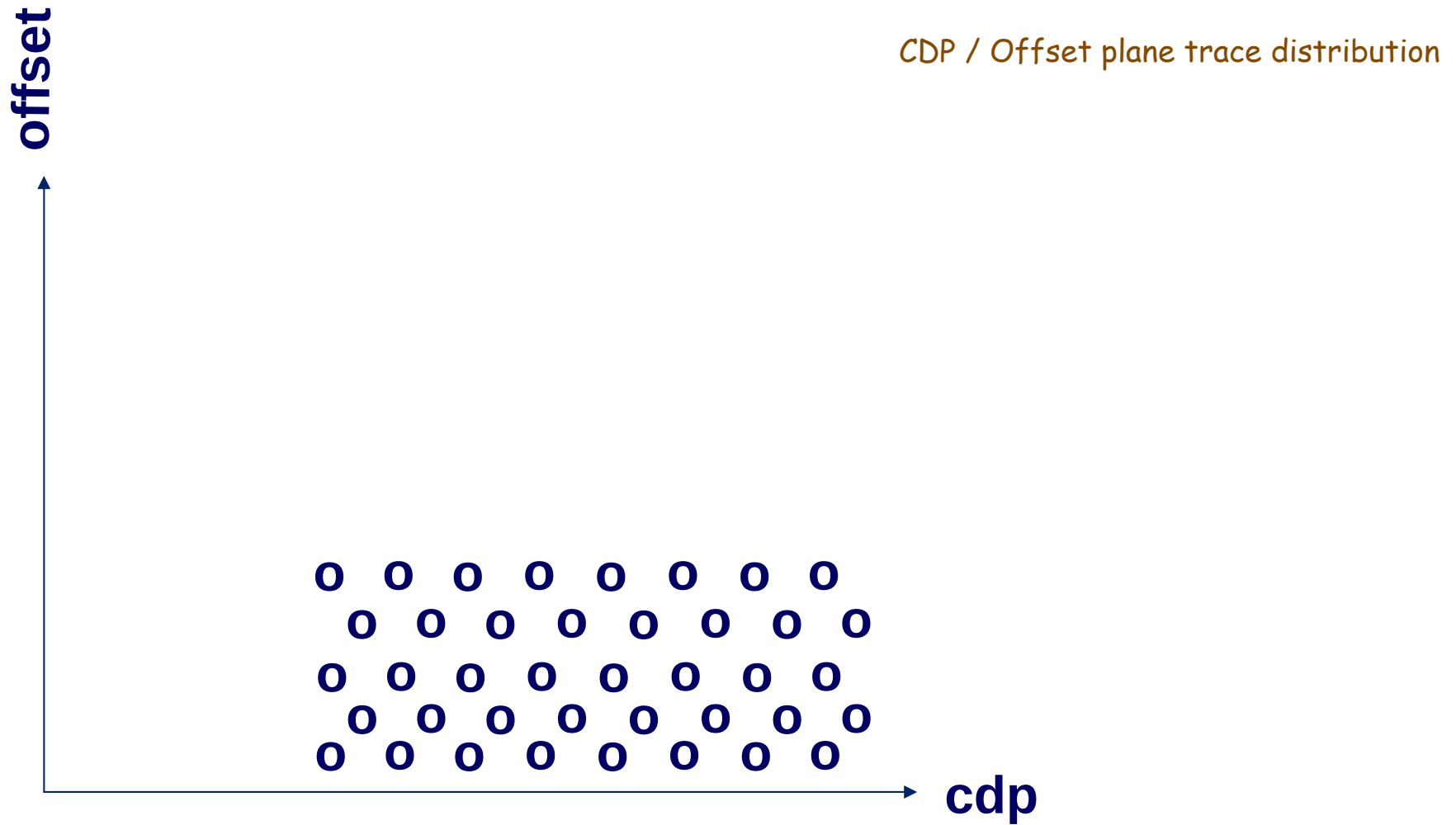
Velocity function stack panel

Introduction to exercises:

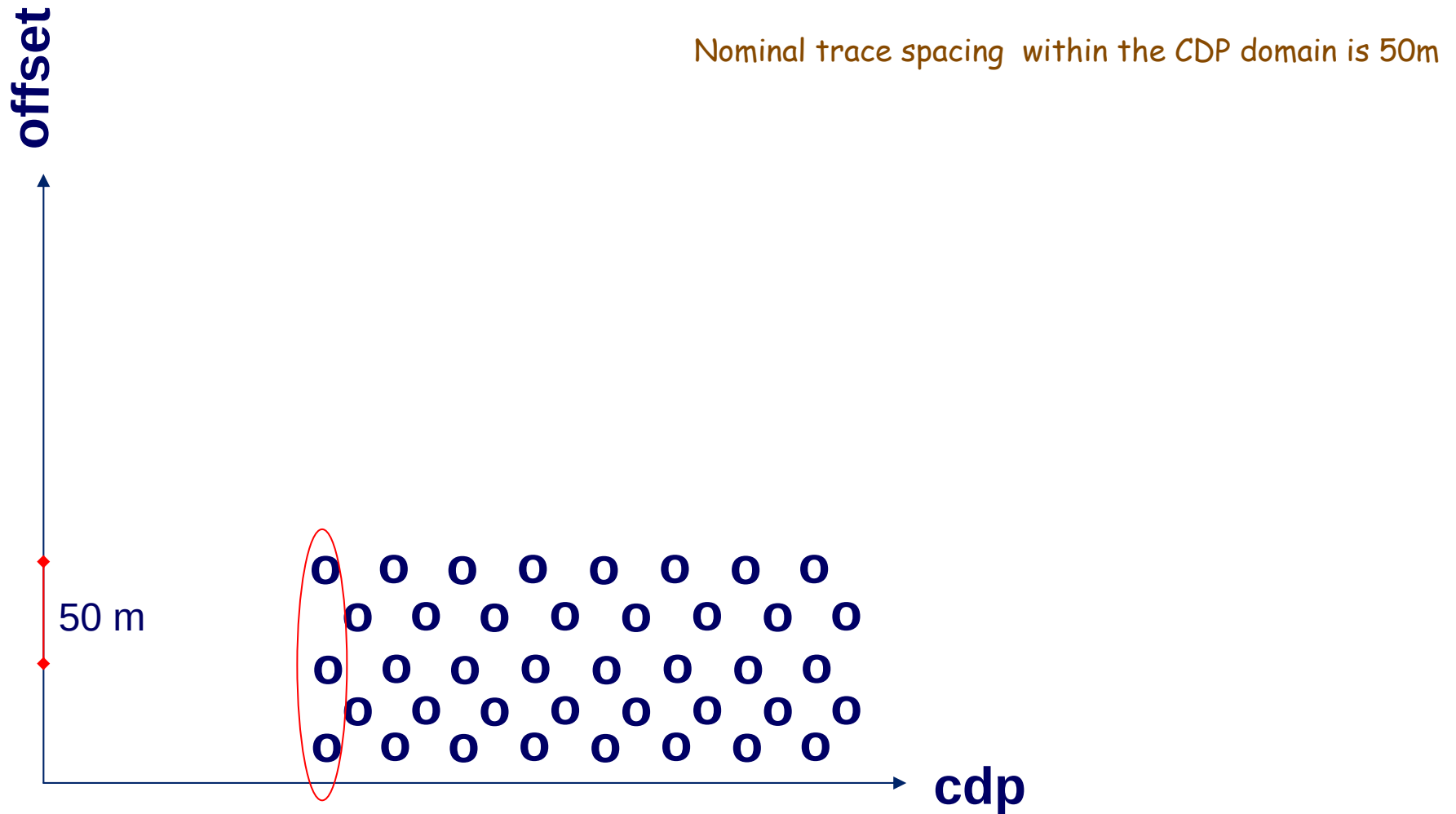
Kirchhoff PreStack Time Migration

Course in
Seismic processing for interpreters

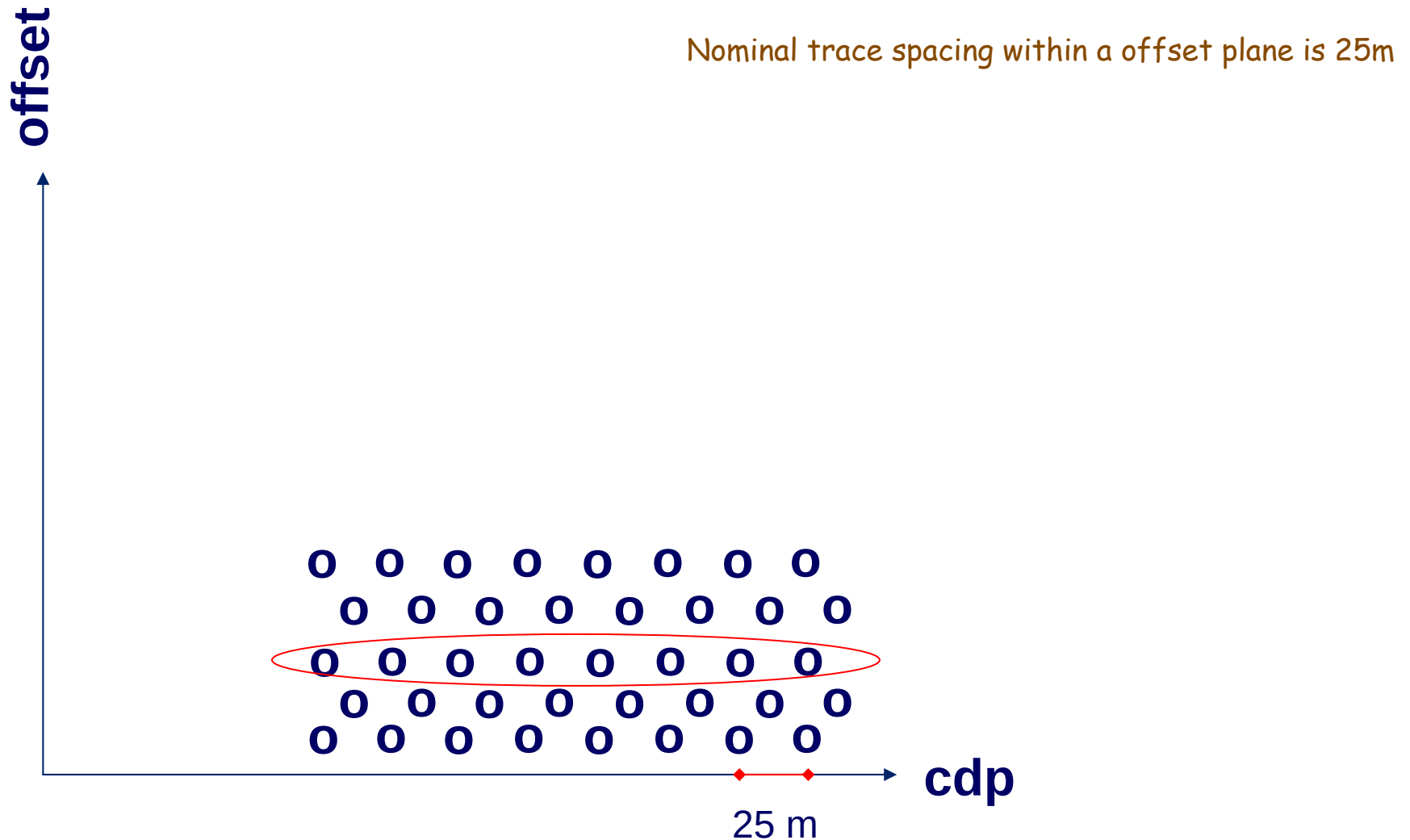
Exercise: Kirchhoff PSTM



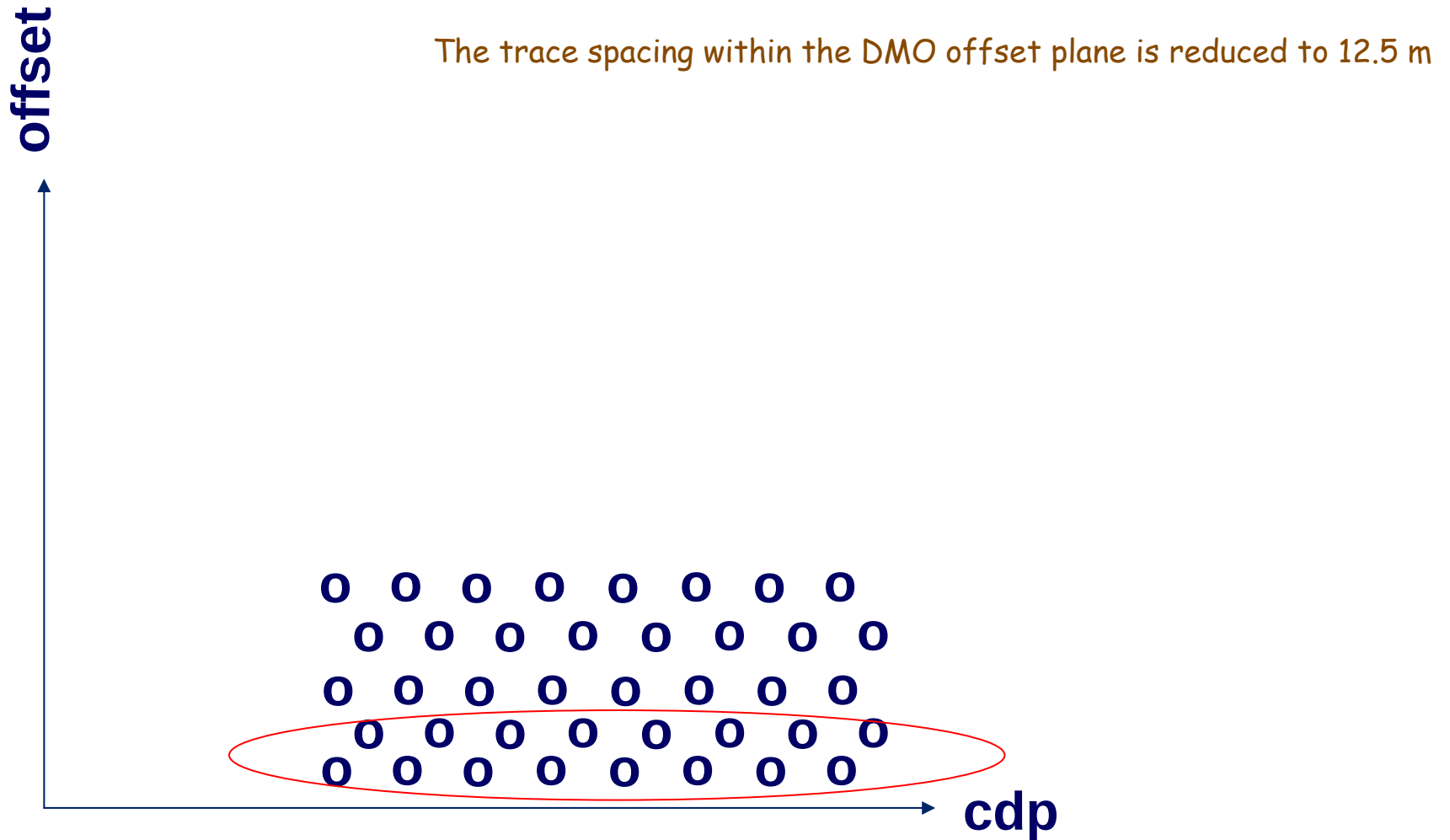
Exercise: Kirchhoff PSTM



Exercise: Kirchhoff PSTM



Exercise: Kirchhoff PSTM



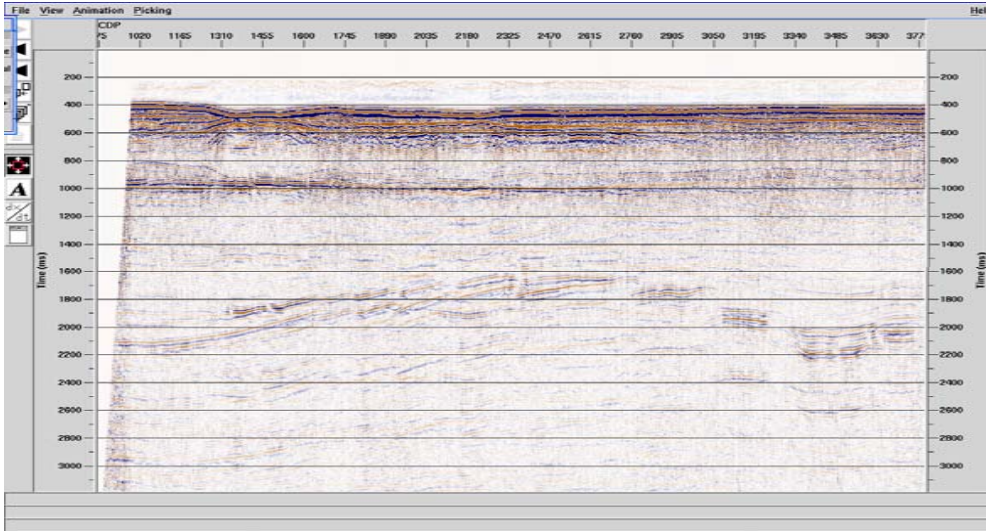
Exercise: Kirchhoff PSTM

Editing Flow: 14) PreStack Time Mig																																		
Add	Delete	Execute View Exit																																
Disk Data Input <- CDPs w/DBS+Radon		Data Input /																																
Common Offset DMO Binning		Disk Data Inp																																
Disk Data Output -> TMP		Disk Data Out																																
Disk Data Input <- TMP		SEG-Y Output																																
Prestack Kirchhoff Time Migration		Well Log Input																																
Normal Moveout Correction		Well Log Input																																
Trace Display Label		Well Log Input																																
Disk Data Output -> K		Well Log Input																																
<table border="1"> <thead> <tr> <th colspan="2">Prestack Kirchhoff Time Mig.</th> </tr> </thead> <tbody> <tr> <td>Perform parallel processing?</td> <td>Yes No</td> </tr> <tr> <td>Number of parallel processes to spawn</td> <td>6</td> </tr> <tr> <td>Specifies the machines to run processes on</td> <td><cpu-1>,<cpu-2>,...,<cpu-6></td> </tr> <tr> <td>CDP interval (feet or meters)</td> <td>12.5</td> </tr> <tr> <td>Explicitly define bounds of output field?</td> <td>Yes No</td> </tr> <tr> <td>First CDP of output field</td> <td>875.</td> </tr> <tr> <td>Last CDP of output field</td> <td>3794.</td> </tr> <tr> <td>CDP spacing of traces in output field</td> <td>1.</td> </tr> <tr> <td>Input a Horizon Table?</td> <td>Yes No</td> </tr> <tr> <td>Maximum frequency (in Hz)</td> <td>60.</td> </tr> <tr> <td>Migration aperture (feet or meters)</td> <td>0.</td> </tr> <tr> <td>Avoid spatial aliasing?</td> <td>Yes No</td> </tr> <tr> <td>Get RMS velocities from database?</td> <td>Yes No</td> </tr> <tr> <td>Select RMS vs. time velocity file</td> <td>Updated Velocity Picks</td> </tr> <tr> <td>Change the default tapering?</td> <td>Yes No</td> </tr> </tbody> </table>			Prestack Kirchhoff Time Mig.		Perform parallel processing?	Yes No	Number of parallel processes to spawn	6	Specifies the machines to run processes on	<cpu-1>,<cpu-2>,...,<cpu-6>	CDP interval (feet or meters)	12.5	Explicitly define bounds of output field?	Yes No	First CDP of output field	875.	Last CDP of output field	3794.	CDP spacing of traces in output field	1.	Input a Horizon Table?	Yes No	Maximum frequency (in Hz)	60.	Migration aperture (feet or meters)	0.	Avoid spatial aliasing?	Yes No	Get RMS velocities from database?	Yes No	Select RMS vs. time velocity file	Updated Velocity Picks	Change the default tapering?	Yes No
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Select RMS vs. time velocity file	Updated Velocity Picks																																	
Change the default tapering?	Yes No																																	

- Try parallel processing
- Use your picked stacking velocities
- Use default value for migration aperture

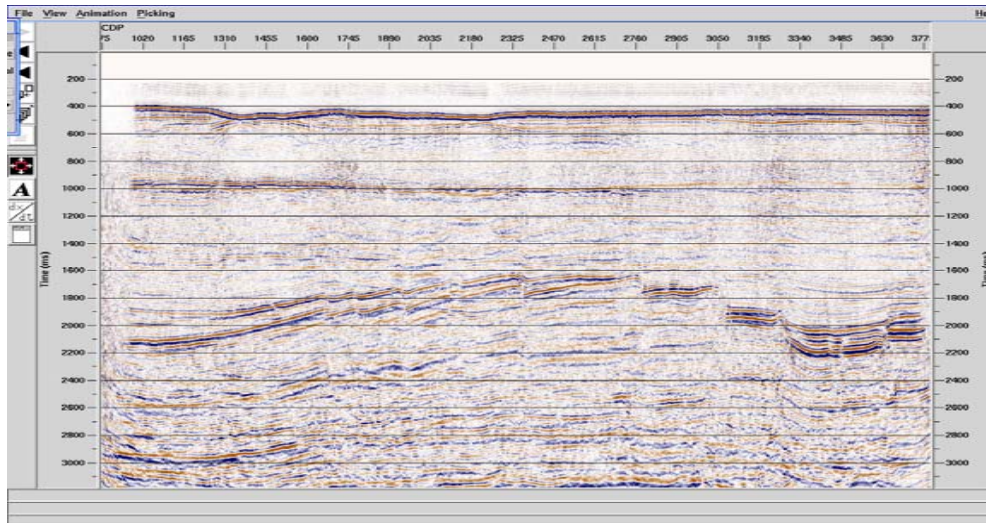
(Normally it is a good idea to avoid spatial aliasing, but we will discard the option here to save runtime)

Exercise: Kirchhoff PSTM



Stack

- Run Kirchhoff PSTM
- Output w/reverse NMO correction.
- Create a PSTM stacked section



Kirchhoff PSTM stack

Introduction to exercises:

PostStack Depth Migration

Course in
Seismic processing for interpreters

Exercise: PostStack Depth Migration

In this exercise:

Build a ProMAX flow that will

- Convert stacking velocities to interval velocities (V_{int}) in depth
- Run a Kirchhoff post-stack depth migration
- View final depth migrated section
- Convert depth-migrated section to time

Compare time-converted depth section to other Snøhvit results

Exercise: PostStack Depth Migration

AREA: Training for Interpreters
LINE: Team A
Editing Flow: 16) PostStack Depth Mig

Add	Delete	Execute	View	Exit	Data Input /
Velocity Manipulation^<					Disk Data In
Disk Data Input < DBS+Radon Stack					Disk Data On
Kirchhoff Depth Mig.					?
Kirchhoff Depth Mig.					
Trace Display Label	CDP interval (feet or meters)	12.5			
Disk Data Output -> K	Maximum frequency to migrate (in Hz)	60.			
Trace Display	Depth sampling interval of output data	0.			
	Maximum depth to migrate	0.			
	Migration aperture (feet or meters)	0.			
	Maximum dip to migrate	180.			
	Avoid spatial aliasing?	Yes No			
	Method for producing the Green's function	Maximum amplitude ray tracing			
	Accuracy factor	1.			
	Get interval velocities from database?	Yes No			
	Select interval vs. time file	IntVels in Depth			
	Velocity smoothing (feet or meters)	100.			
	Change maximum memory usage?	Yes No			
	Change the default tapering?	Yes No			
	Re-apply trace mutes in depth?	Yes No			
	Re-kill dead traces?	Yes No			

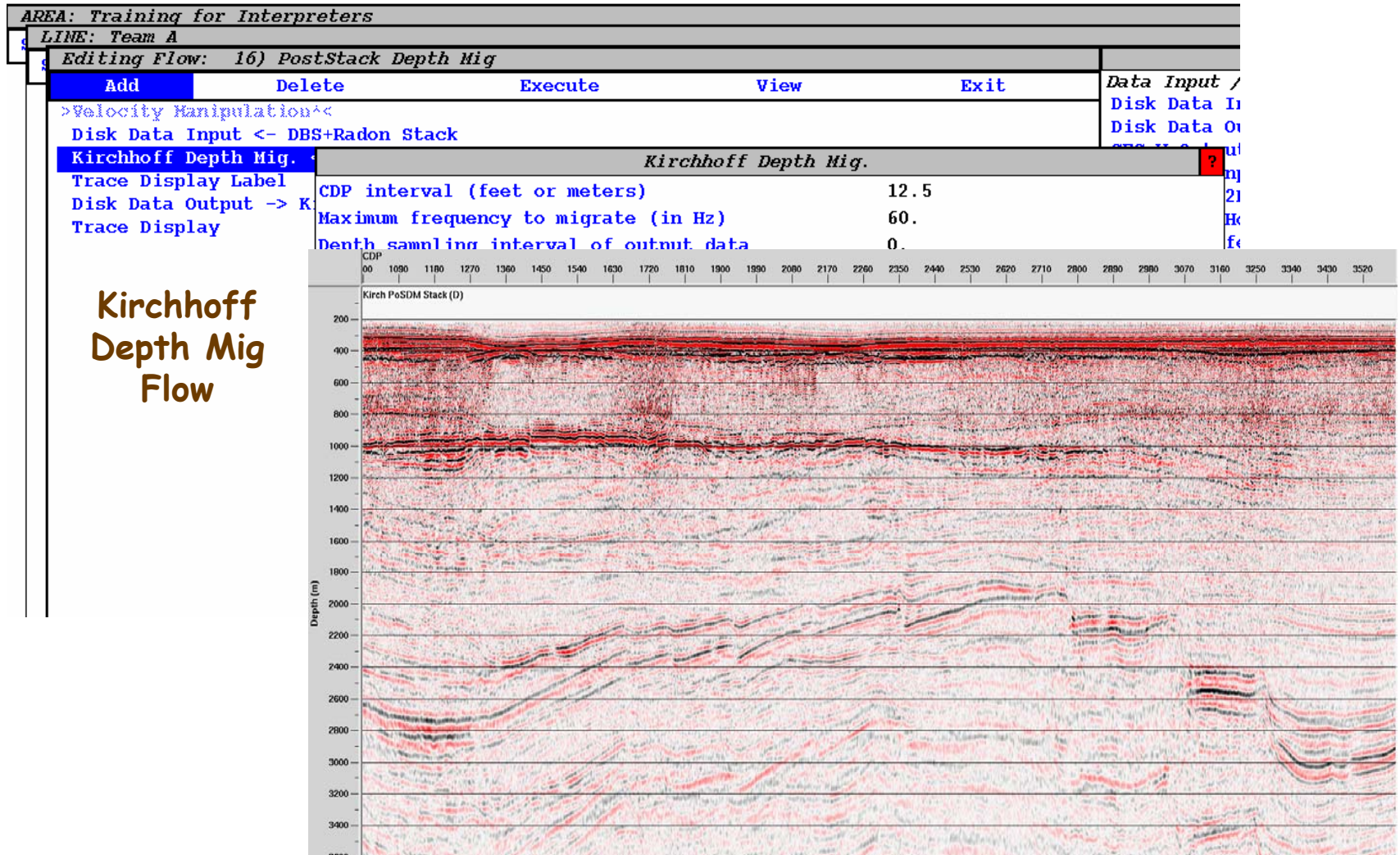
Pad Traces

Kirchhoff
Depth Mig
Flow

1) Convert stacking velocities to interval velocities in depth

2) Use *IntVels in Depth* in the Kirchhoff Depth Migration

Exercise: PostStack Depth Migration

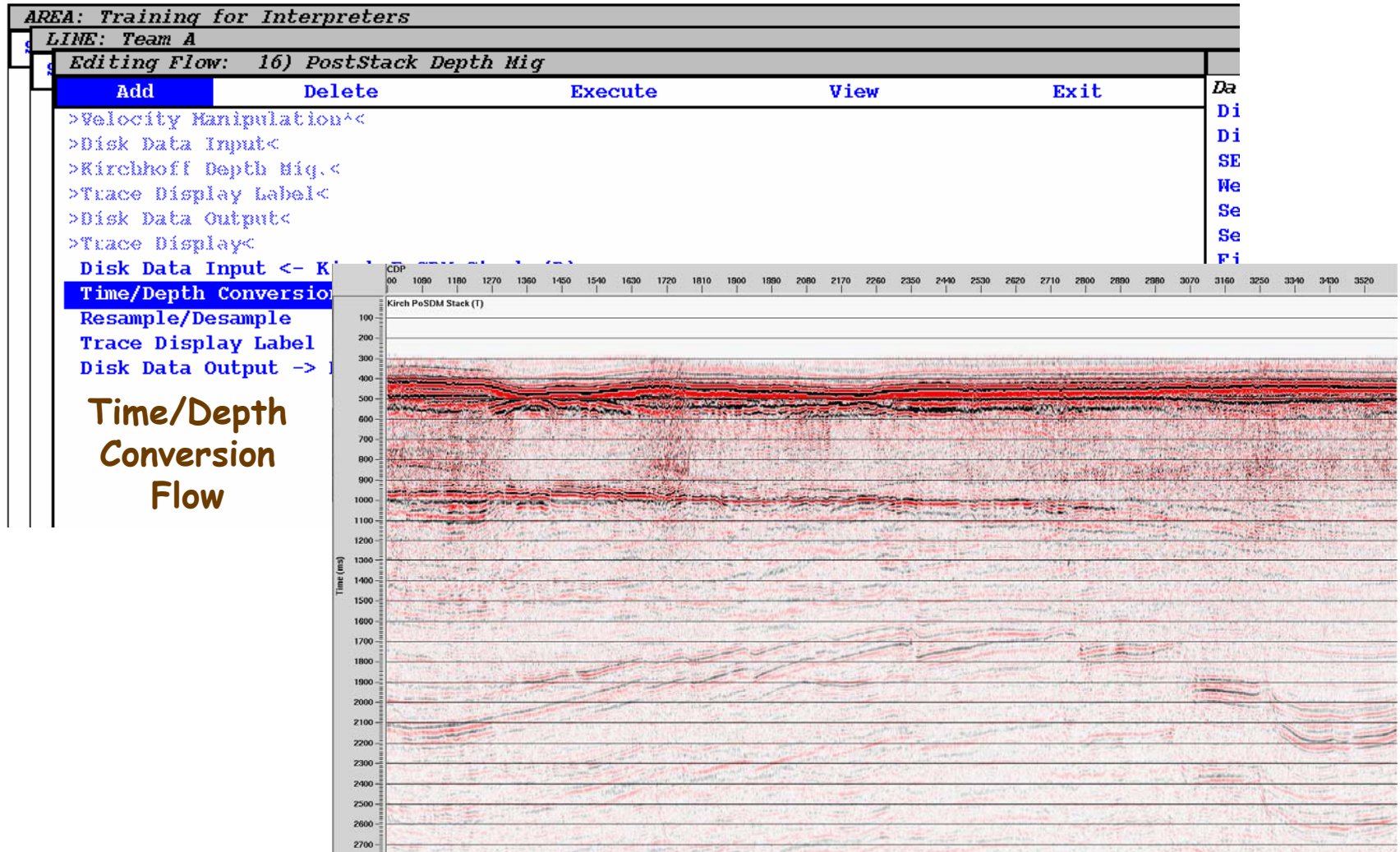


Kirchhoff Depth Mig View

Exercise: PostStack Depth Migration

AREA: Training for Interpreters		
LINE: Team A		
Editing Flow: 16) PostStack Depth Mig		
Add	Delete	Execute View Exit
>Velocity Manipulation^< >Disk Data Input< >Kirchhoff Depth Mig.< >Trace Display Label< >Disk Data Output< >Trace Display< Disk Data Input <- Kirch PoSDM Stack (D)		
Time/Depth Conversion	Time/Depth Conversion	
Resample/Desample	Conversion direction Depth-to-TIME	
Trace Display Label	Maximum frequency of interest (in Hz)	60.
Disk Data Output -> K	Percent velocity scale factor	100.
Time/Depth Conversion Flow	Type of velocity table to use	Interval Velocity in depth
	Get velocities from DATABASE?	Yes No
	SELECT Velocity Parameter File	IntVels in Depth
	Convert Mutes?	Yes No

Exercise: PostStack Depth Migration



Time-converted Kirchhoff Depth Mig View